



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



AGENTIC AI

Bridging the Real Physical Gap -- Multimodal Data Synthesis and Closed-Loop Simulation of World Models in Embodied Intelligence

Talk by: Zhen Li Associate Professor (Tenured)

Presenter: Dr. Xiaoyu Ye

FNii-Shenzhen, the Chinese University of Hong Kong, Shenzhen (CUHKSZ)

July 25, 2026, Shenzhen

L2-L4 level Autonomous Driving



Tesla AutoPolit



RoboTaxi

World Model for Autonomous Driving



Wayve GAIA-1



Wayve GAIA-2

<https://wayve.ai/thinking/gaia-2/>

Outline

Robust and Generalized 3D Understanding for L3/L4 AD and Embodied AI

- ◆ Multimodal 3D Data
 - Paired 3D-Text Native Data and Benchmark
- ◆ Pre-training
 - Vision Pre-training for autonomous Driving
- ◆ Robust and Generalized 3D Perception
 - Robust Perception for OOD Scenarios

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Robust and Generalized 3D Understanding for L3/L4 AD and Embodied AI

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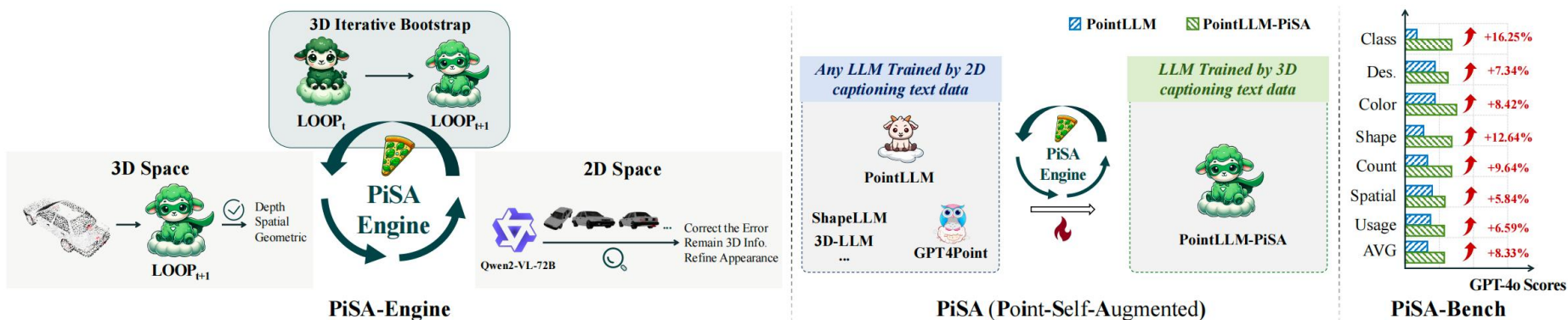


PiSA: A Self-Augmented Data Engine and Training Strategy for 3D Understanding with Large Models

Zilu Guo^{1,2}, Hongbin Lin^{1,2}, Zhihao Yuan^{1,2}, Chaoda Zheng^{1,2}, Pengshuo Qiu³
Dongzhi Jiang³, Renrui Zhang³, Chun-Mei Feng⁴, Zhen Li^{2,1}✉

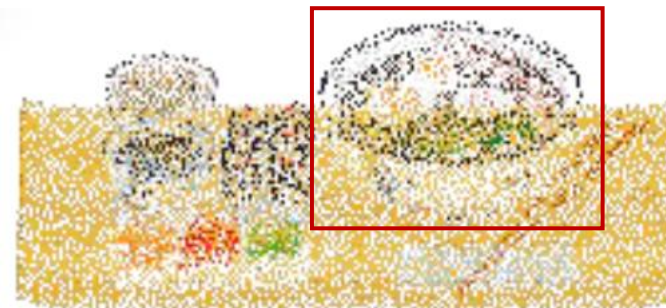
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Challenges of Current 3D MLLMs:

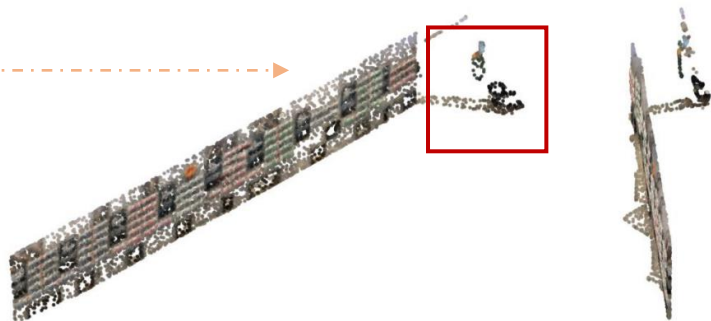
- Wrong Comprehension
- Wrong Attribute
- Missing Object



Missing Noodles!

Q: What is this?

A: This is sushi. It is made from vinegared rice, often served with raw fish or other ingredients, wrapped in seaweed or wrapped in paper.



Q: What is the shape of the 3D object model?

A: The shape of the 3D object model is a **cuboid**.



Q: What is the color of the 3D model airplane?

A: The airplane is **predominantly white in color**.



Motivations

Why does the 3D MLLM often fail to universally understand 3D object?

Our analysis : **Unqualified/Lack 3D instruction data**

Training data → captioning is wrong and full with hallucination → without 3D knowledge

Ground Truth	A storage box		 Inaccuracy and Hallucination
Cap3D	A dark brown and light brown truck		
Prompt	Caption this 3D model in detail.		
Point-Bind LLM	The image features a 3D model of a house with a box sitting on the roof. The box is placed on top of a small house, which is positioned on top of a larger house. The larger house has a window and a door, while the smaller house has a window. The scene appears to be a playful representation of a house with a box on top, possibly for a game or an artistic project. The 3D model is a creative rendition of a box designed to appear like a small coffin. The box has an intriguing dark color, perhaps meant to mimic the dark hue of a traditional coffin. It's covered with a light pink sheet, adding a whimsical and unexpected contrast to the overall dark tone. Due to the design, the box might be intended for containing small items like jewelry or trinkets, and the pink sheet could be a protective layer or simply aesthetic fluff.		
PointLLM			





Motivations

Why does current 3D benchmark fail to specifically evaluate 3D tasks?

Our analysis : **Evaluated attributes are not enough**

 Description	 Color	 Shape	 Usage	 Class
PiSA-Bench: This 3D model displays a wooden rectangular platform adorned with various items on top. <i>The setup</i> features a sizable dark clay vessel or cauldron positioned on a miniature stand at one end of the platform. <i>At the other end</i>, a table-like structure supports books, miniature vases, and what seems to be a witch's hat. <i>The entire arrangement</i> is decorated with vibrant speckles or splashes of paint in red, green, and blue, creating a magical or whimsical feel. <i>The wood surface</i> shows clear plank marks and a textured finish. Cap3D: A wooden rectangular board with a clay pot on a three stand and a table having some utensils on top. X	PiSA-Bench: The base displays a wooden brown shade, decorated with vibrant speckles in red, green, and blue. The large vessel is dark gray. The table holds brown books, two small vases (one blue and one green), and a dark gray pointed hat. The table also features the same speckled pattern as the wooden platform. Cap3D: None X	PiSA-Bench: The principal structure is a rectangular wooden platform. It includes a circular pot at one end and a rectangular table at the other. Cap3D: ...rectangular board... X	PiSA-Bench: This 3D model can be used in video games, animations, or virtual settings to craft a scene with a magical or fantasy theme, perhaps associated with witchcraft or alchemy. Cap3D: None X	PiSA-Bench: "wooden board", "table", "pottery" <i>Prompt: refer to their subclass or one of answers from label</i> Cap3D: ...board...pot...table <i>GPT: F. #Reason: The first refers to a board and pot, the second to a spherical bowl.</i> X
			 Spatial	 Count
			PiSA-Bench: The clay vessel is situated at one end of the wooden platform. The table, bearing various items, is located at the opposite end. All elements are positioned on the wooden platform. Cap3D: ...on top... X	PiSA-Bench: The arrangement includes several items: one large pot, one table, multiple books, two small vases, and one pointed hat. Cap3D: A...some X

We propose **PiSA-Bench**, a comprehensive 3D evaluation benchmark encompassing seven aspects of attributes.



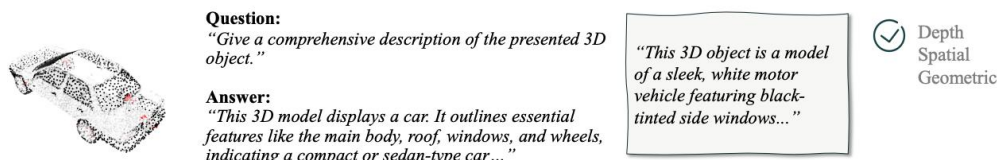
Method

Question: How to make the model learn enough 3D knowledge?

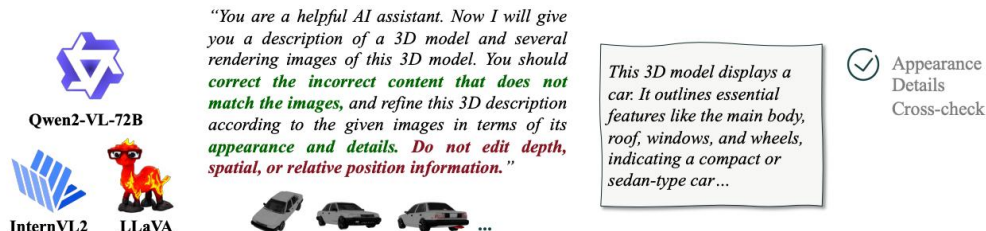
Idea: **Using native 3D instruction data for training**

➤ A framework for generating instruction point-language datasets enriched with 3D spatial semantics.

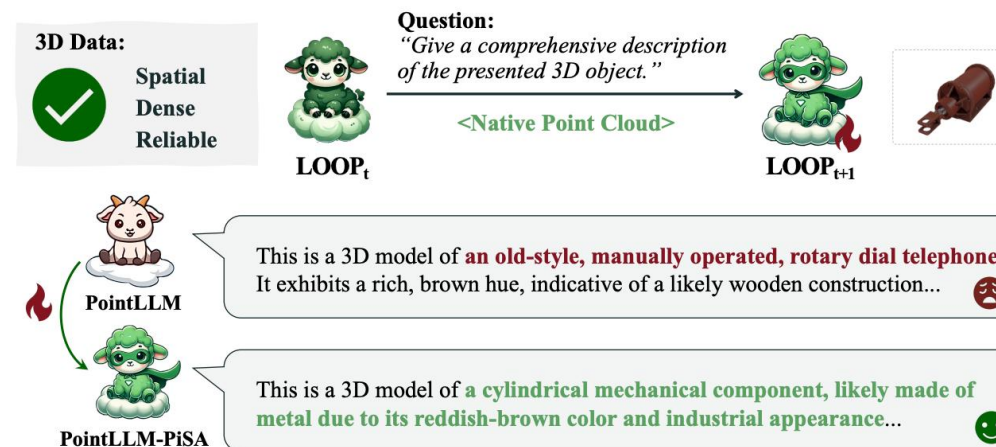
Stage 1: 3D-space Data Annotation



Stage 2: 2D-space Data Refinement



Stage 3: Iterative 3D Data Bootstrap





Experiments

Quantitative Results on 3D captioning Benchmark

- Our PiSA-Bench

- Human evaluation
- GPT-4o evaluation
- Traditional metrics

Model	Input	Desc.	Color	Shape	Count	Spatial	Usage	AVG
LLaVA-7B [16]	Single-V. Img.	23.04	20.50	28.21	40.58	1.42	22.18	22.66
LLaVA-13B [16]	Single-V. Img.	24.27	18.97	25.16	45.42	1.89	23.03	23.12
3D-LLM [10]	3D Obj. + Mul.-V. Img.	15.75	11.96	19.39	45.03	2.54	22.80	19.58
ShapeLLM-7B [22]	3D Point Cloud	22.47	12.57	20.45	49.21	2.13	17.94	20.80
ShapeLLM-13B	3D Point Cloud	25.13	21.67	28.28	50.90	0.13	19.49	24.27
Point-Bind LLM-7B	3D Point Cloud	1.54	0.00	2.31	3.08	0.00	1.92	1.48
PointLLM-7B	3D Point Cloud	35.42	39.74	36.47	61.67	2.91	52.48	38.12
PointLLM-PiSA-7B	3D Point Cloud	39.72 (↑ 4.30)	46.44	48.31	65.74	5.84	56.15	43.70 (↑ 5.58)
PointLLM-PiSA²-7B	3D Point Cloud	42.64 (↑ 7.22)	48.05	49.11	71.31	8.67	58.66	46.41 (↑ 8.29)
PointLLM-PiSA³-7B	3D Point Cloud	42.76 (↑ 7.34)	48.16	48.88	71.05	8.78	59.07	46.45 (↑ 8.33)
PointLLM-PiSA⁴-7B	3D Point Cloud	42.76	47.78	48.91	71.13	8.65	57.99	46.20
PointLLM-13B	3D Point Cloud	37.44	42.69	46.15	62.18	8.33	49.96	41.13
PointLLM-PiSA-13B	3D Point Cloud	42.95 (↑ 5.51)	49.87	52.31	73.08	9.62	53.64	46.91 (↑ 5.78)

Model	GPT-4o	Sentence-BERT	SimCSE	Corr.↑
ShapeLLM-7B	28.04	41.61	42.18	2.49
Point-Bind LLM-7B	2.10	26.54	24.66	0.14
PointLLM-7B	28.41	46.59	47.18	4.56
PointLLM-PiSA-7B	44.87	48.21	48.38	5.78
PointLLM-PiSA²-7B	47.91	48.66	49.04	5.94
ShapeLLM-13B	32.52	42.12	43.03	3.44
PointLLM-13B	33.69	48.90	50.21	4.76
PointLLM-PiSA-13B	47.45	49.78	50.09	6.32



Experiments

Quantitative Results on 3D generative classification Benchmark

- Objaverse

Model	Input	Objaverse (I)	Objaverse (C)	Average
LLaVA-7B [16]	Single-V. Img.	31.50	33.50	32.50
LLaVA-13B [16]	Single-V. Img.	34.00	36.50	35.25
3D-LLM [10]	3D Obj. + Mul.-V. Img.	32.00	27.50	29.75
ShapeLLM-7B [22]	3D Point Cloud	41.00	34.00	37.50
ShapeLLM-13B [22]	3D Point Cloud	44.50	43.00	43.75
Point-Bind LLM-7B	3D Point Cloud	1.00	0.50	0.75
PointLLM-7B [32]	3D Point Cloud	39.00	36.00	37.50
PointLLM-PiSA-7B	3D Point Cloud	45.00	46.50	45.75
PointLLM-PiSA²-7B	3D Point Cloud	46.00	48.00	47.00
PointLLM-PiSA³-7B	3D Point Cloud	46.50	47.50	47.00
PointLLM-13B [32]	3D Point Cloud	37.50	40.0	38.75
PointLLM-PiSA-13B	3D Point Cloud	51.50	49.50	50.50
PointLLM-PiSA-13B*	3D Point Cloud	53.50	50.50	52.00

- PiSA-Bench

Model	Input	PiSA-Bench (I)	PiSA-Bench (C)	Average
3D-LLM	3D Obj. + Mul.-V. Img.	20.00	32.50	26.75
ShapeLLM-7B	3D Point Cloud	27.50	37.50	32.50
ShapeLLM-13B	3D Point Cloud	55.00	42.50	48.75
Point-Bind LLM-7B	3D Point Cloud	7.50	5.00	6.25
PointLLM-7B	3D Point Cloud	47.50	47.50	47.50
PointLLM-PiSA-7B	3D Point Cloud	60.00	62.50	61.25 (↑ 13.75)
PointLLM-PiSA²-7B	3D Point Cloud	62.50	62.50	62.50 (↑ 15.00)
PointLLM-PiSA³-7B	3D Point Cloud	60.00	67.50	63.75 (↑ 16.25)
PointLLM-13B	3D Point Cloud	52.50	55.00	53.75
PointLLM-PiSA-13B	3D Point Cloud	55.00	57.50	56.25 (↑ 2.50)
PointLLM-PiSA-13B*	3D Point Cloud	67.50	65.00	66.25 (↑ 12.50)

- Instruction-typed (I) prompt “What is this?” and the Completion-typed (C) prompt “This is an object of...”

*indicates those models trained with 182K data.

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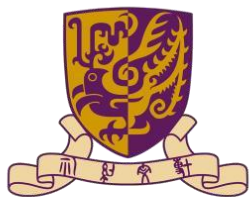
- Paired 3D-Text Native Data and Benchmark

- ◆ Pre-training

- Vision Pre-training for autonomous Driving

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未来智联网络研究院



VisionPAD: A Vision-Centric Pre-training Paradigm for Autonomous Driving

*Haiming Zhang^{1,2}, Wending Zhou^{1,2}, Yiyao Zhu³, Xu Yan^{4†}, Jiantao Gao⁴,
Dongfeng Bai⁴, Yingjie Cai⁴, Bingbing Liu⁴, Shuguang Cui^{2,1}, Zhen Li^{2,1†}*

¹ The Future Network of Intelligence Institute, The Chinese University of Hong Kong (Shenzhen),

² School of Science and Engineering, The Chinese University of Hong Kong (Shenzhen),

³ HKUST

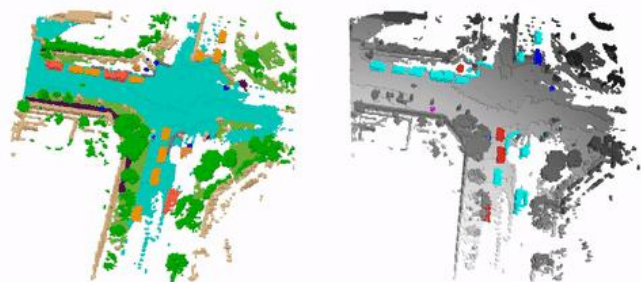
⁴ Huawei Noah's Ark Lab

Vision-centric 3D Perception Tasks:

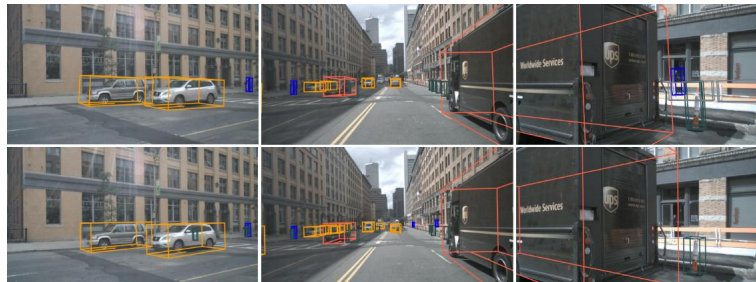
- Inputs: Multi-view camera images
- Outputs: 3D bounding boxes (3D object detection), 3D semantic occupancy, map segmentation
- Advantages: cost-effectiveness, general object representation, suitable for unified models



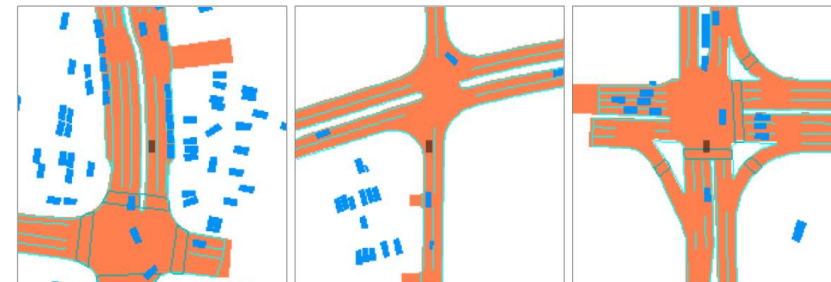
Multi-view images



3D semantic occupancy prediction



3D object detection

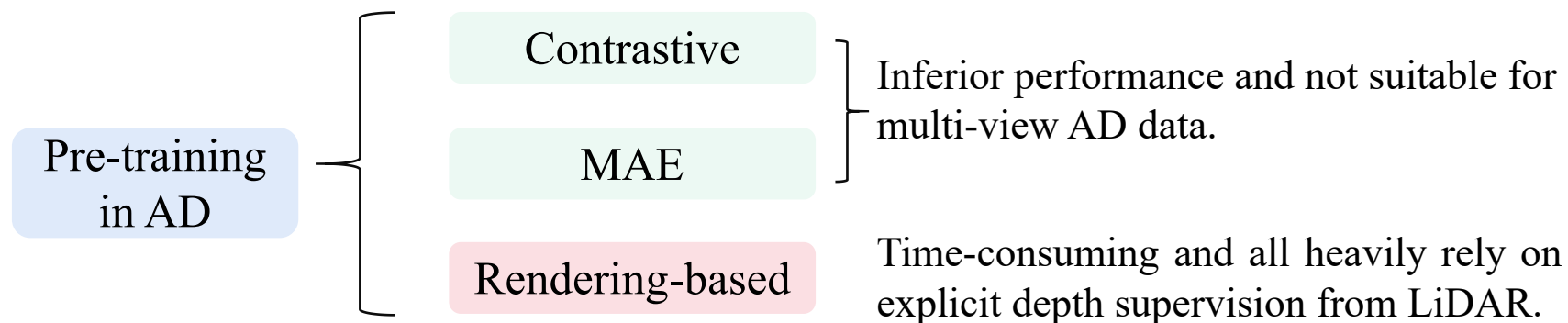


Map segmentation

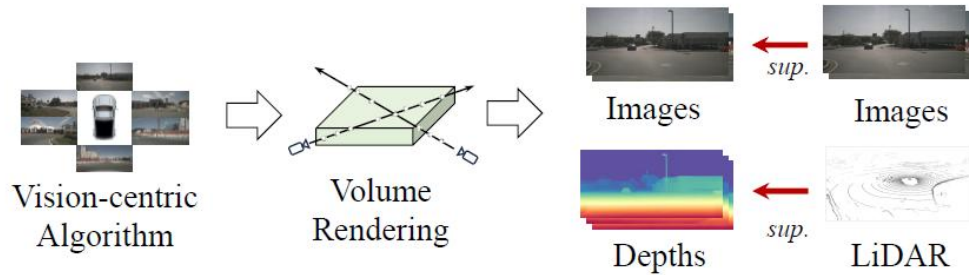
Challenges of Scaling up Vision-centric Perception Models:

- Lack of large-scale 3D annotations;
- Time-consuming and high-load to train models with large parameters;

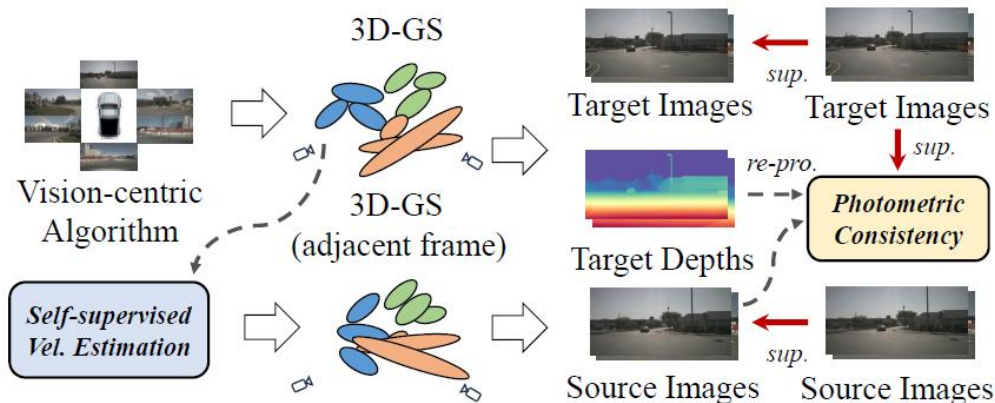
How to solve? Pre-training is an effective method to rescue.



Could we design an efficient self-supervised pre-training paradigm solely rely on vision inputs?



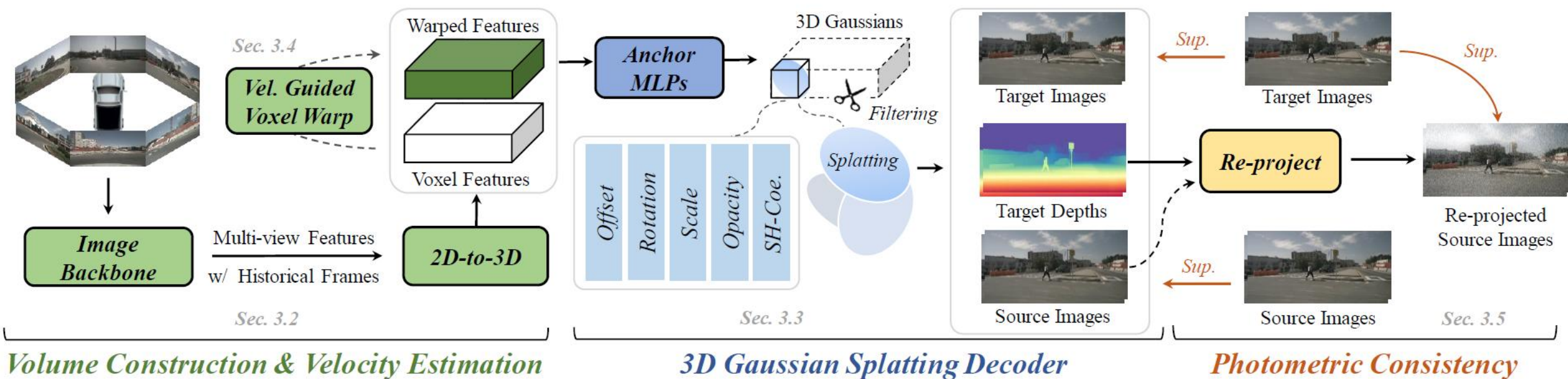
(a) UniPAD: volume rendering with explicit depth supervision



(b) VisionPAD: 3D-GS with solely vision-centric supervision

- First introduce a more efficient **anchor-based 3DGS representation** in vision-centric perception models ;
- Propose a **photometric consistency** module to impose geometric information into the volume feature **without utilizing the LiDAR sensors**;
- The **self-supervised volume velocity estimation module** further enhance the motion cues.

Framework



- Any vision-centric perception models build the volumetric features;
- An efficient anchor-based 3DGS representation built upon the volume feature by shallow MLPs;
- Photometric consistency loss and self-supervised velocity estimation modules ensure the pre-training performances;

Datasets and Tasks

- nuScenes: 3D object detection, map segmentation
- nuScenes-Occ3D: 3D semantic occupancy prediction

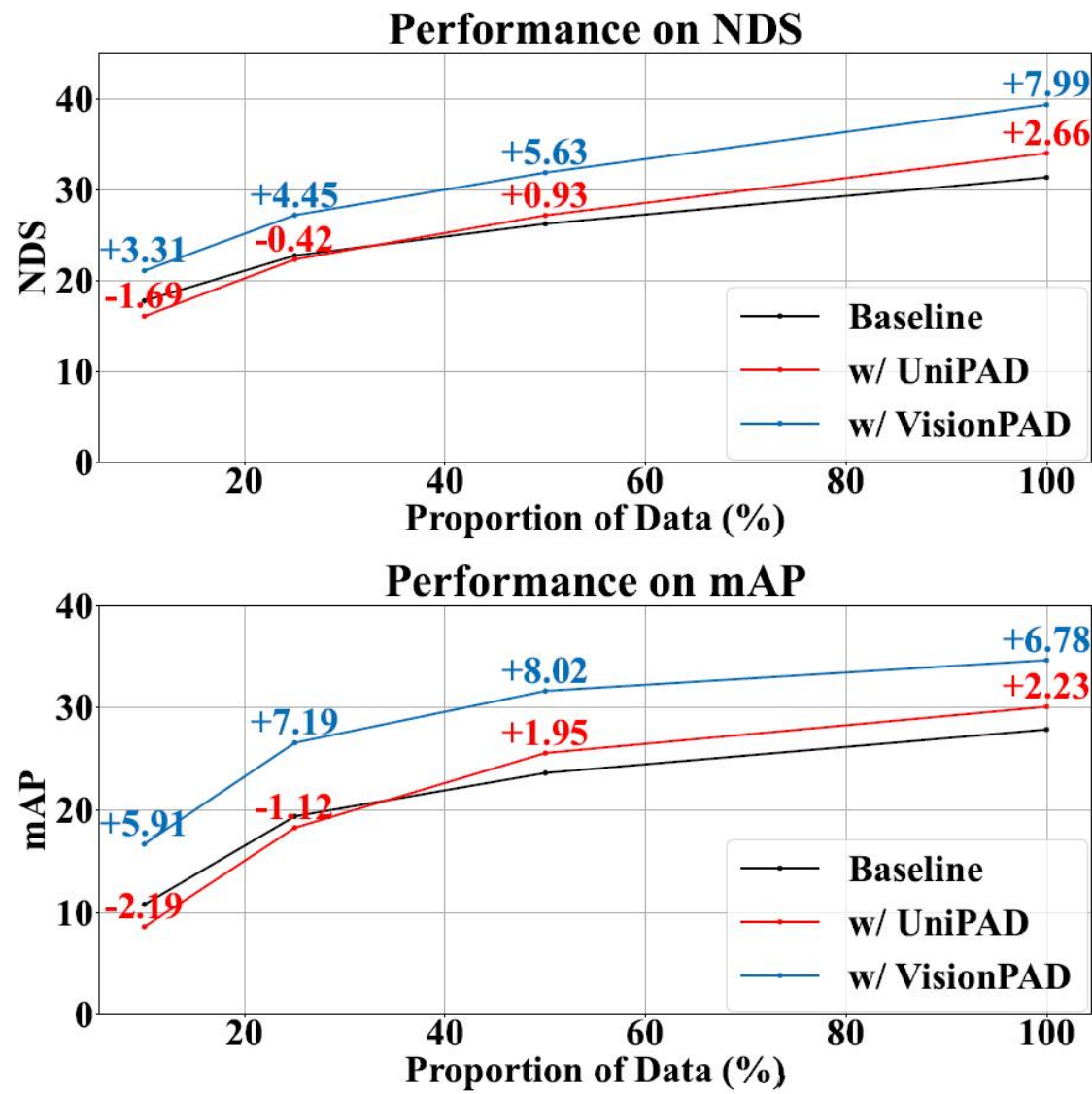
Metrics

- **3D object detection:** nuScenes Detection Score (NDS) and mean Average Precision (mAP)
- **3D occupancy prediction:** mean Intersection-over-Union (mIoU);
- **Map segmentation:** Intersection-over-Union (IoU)

Finetune Settings

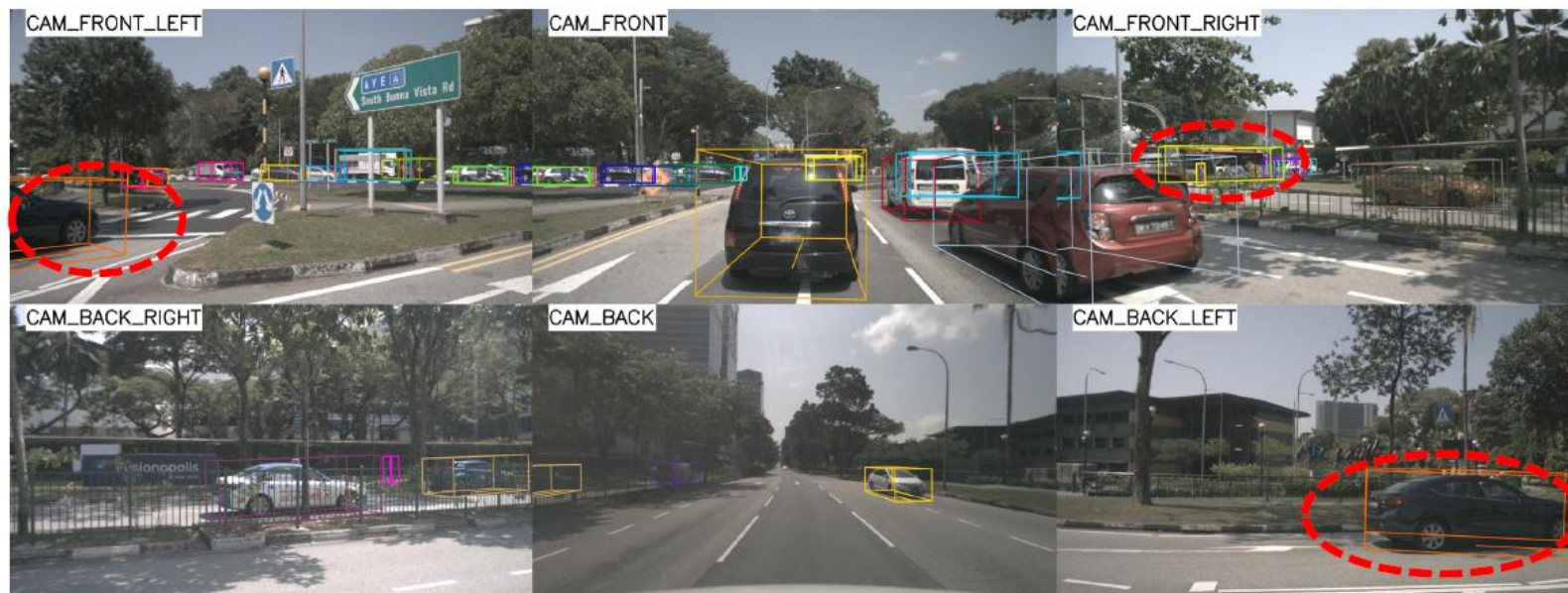
- We strictly follow the official training configurations during fine-tuning without any modifications.

Data efficiency analysis with limited data

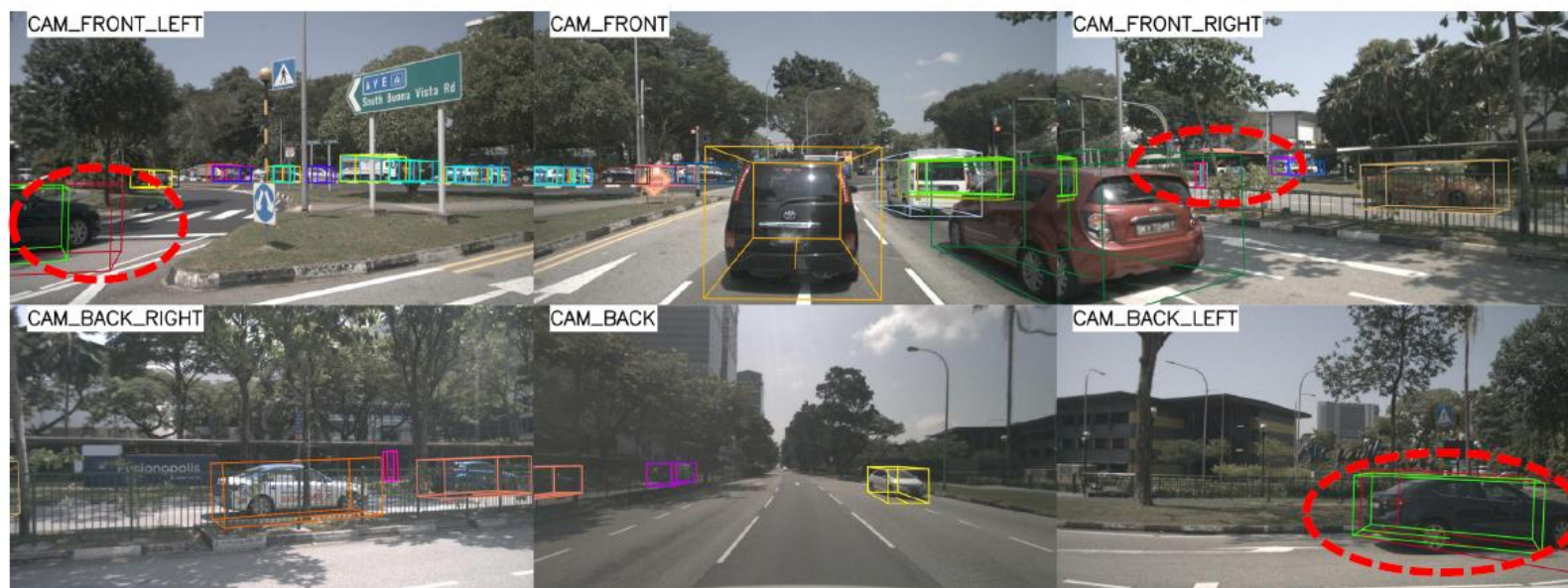


Visualization

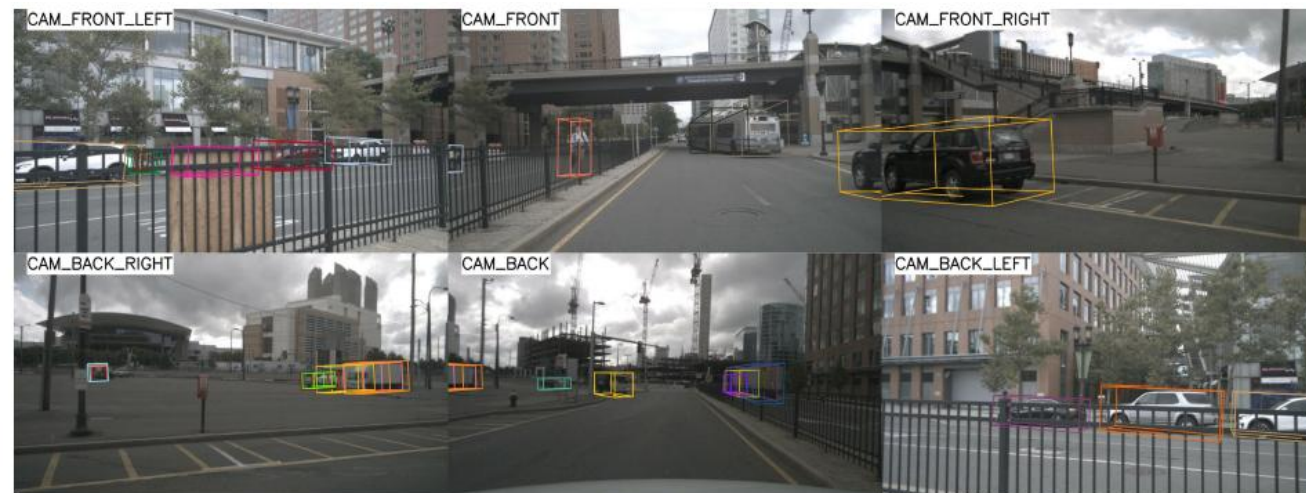
UniPAD



Ours



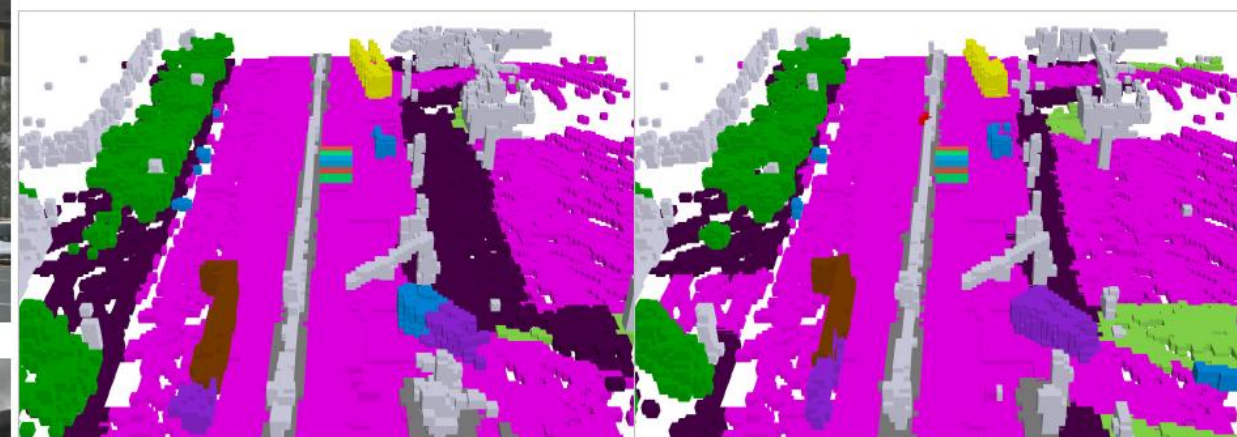
Visualization

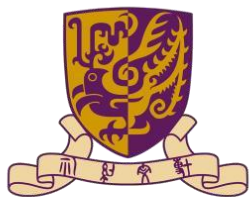


3D Object Detection



3D Object Detection Ground Truth





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SQS: Enhancing Sparse Perception Models via Query-based Splatting in Autonomous Driving

Haiming Zhang^{1,2}, Yiyao Zhu³, Wending Zhou^{1,2}, Xu Yan⁴†✉

Yingjie Cai⁴, Bingbing Liu⁴, Shuguang Cui^{2,1}, Zhen Li^{2,1}†✉

(NeurIPS 2025 Spotlight)

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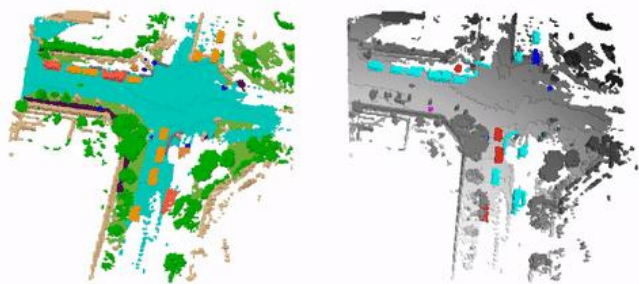
Background

Vision-centric 3D Perception Tasks:

- **Inputs:** Multi-view camera images
- **Outputs:** 3D bounding boxes (3D object detection), 3D semantic occupancy, map segmentation



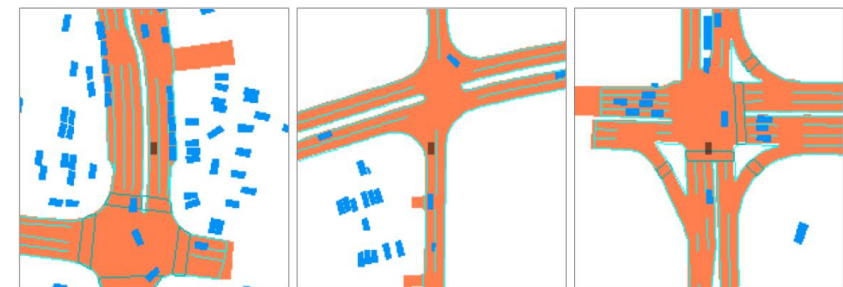
Multi-view images



3D semantic occupancy prediction

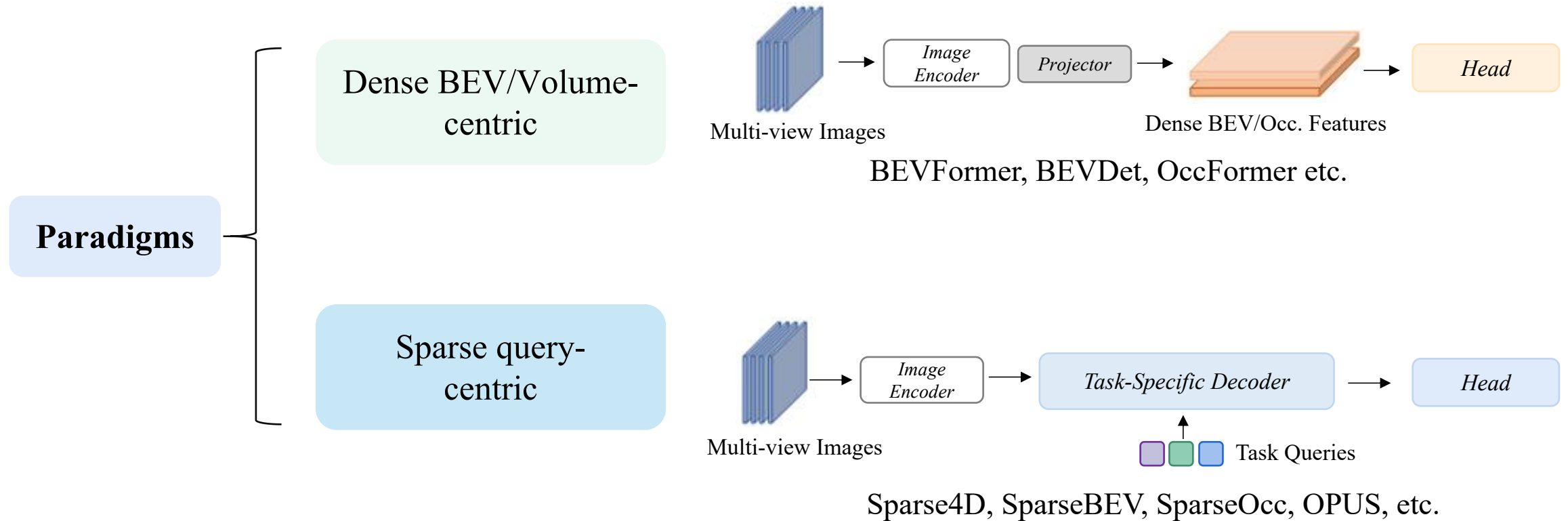


3D object detection



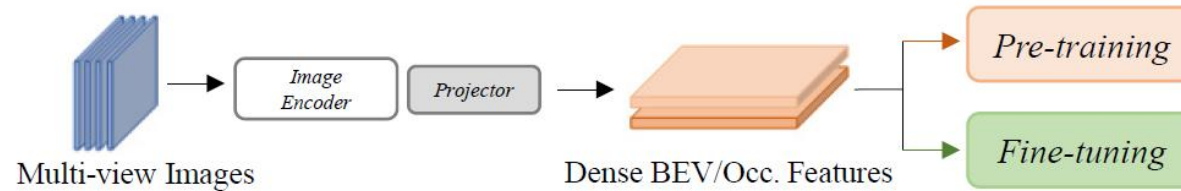
Map segmentation

Existing Vision-centric 3D Perception Paradigms:

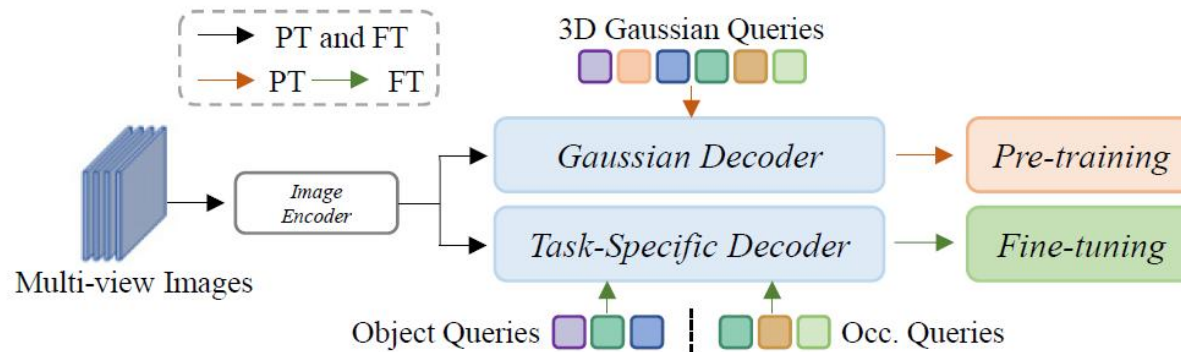


Motivation

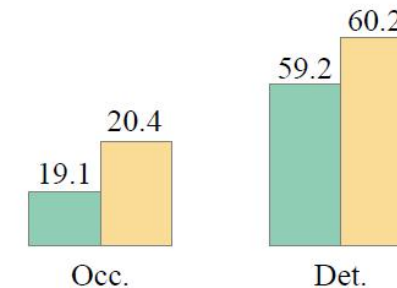
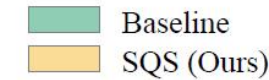
Could we design a unified self-supervised pre-training paradigm to enhance different sparse perception model (SPMs)?



(a) Existing dense BEV/Occ-Centric pre-training and fine-tuning paradigm



(b) Sparse query-based pre-training and fine-tuning paradigm

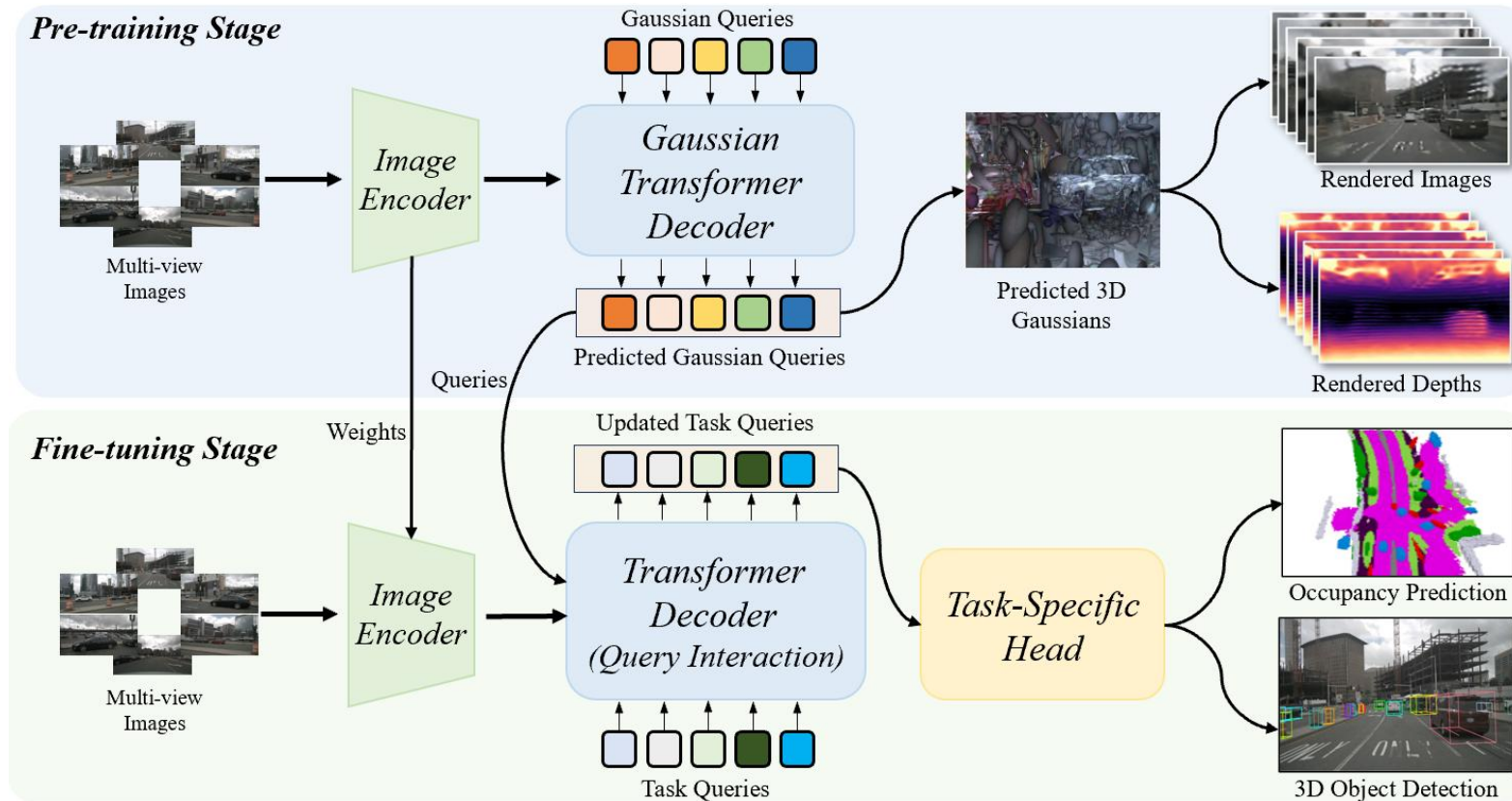


(c) The performance improvements after pre-training by our proposed approach

- The proposed **SQS (Sparse Query-based Splatting)** can be integrated into **any sparse query-based perception model**, accepting Gaussian queries for pre-training and utilizing them for prediction;
- We demonstrate the **effectiveness** of SQS on **query-based 3D semantic occupancy prediction (Occ.)** and **3D object detection (Det.)** tasks.

Sparse Query-based Splatting (SQS)

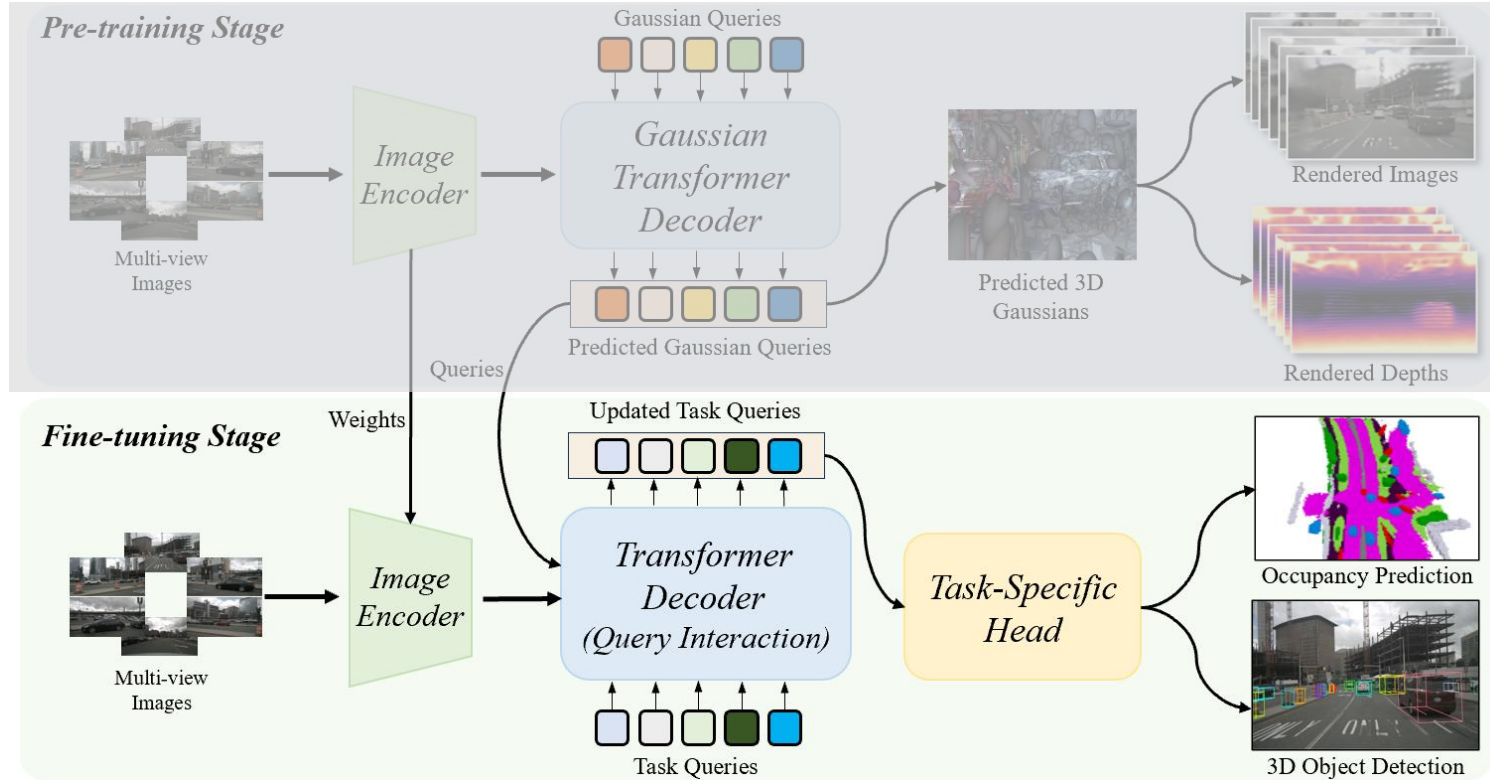
Framework



- A **sparse query-based 3D Gaussian Splatting pre-training** paradigm with RGB image and depth as supervision;
- A **query interaction module** to fully exploit the knowledge encapsulated in the pre-trained queries;
- The light-weight pre-training paradigm can be **plugged** into **any** sparse query-based downstream tasks to enhance their performance.

Sparse Query-based Splatting (SQS)

Query Interaction for Fine-tuning



$$q_t = \text{LocalAttn}(q_t + \text{MLP}(\mu_t), q_k + \text{MLP}(g_k)).$$

- The queries in different SPMs play various roles, so it's hard to share the queries in the pre-training stage;
- Therefore, we propose a plug-in framework based on Query Interaction operation;

Datasets and Tasks

- NuScenes: 3D object detection task
- Dense occupancy annotations from SurroundOcc: 3D semantic occupancy prediction task

Metrics

- **3D object detection:** NuScenes Detection Score (NDS) and mean Average Precision (mAP)
- **3D occupancy prediction:** mean Intersection-over-Union (mIoU), Intersection-over-Union (IoU);

Finetune Settings

- We strictly follow the official training configurations during fine-tuning without any modifications.

Data efficiency analysis with limited data

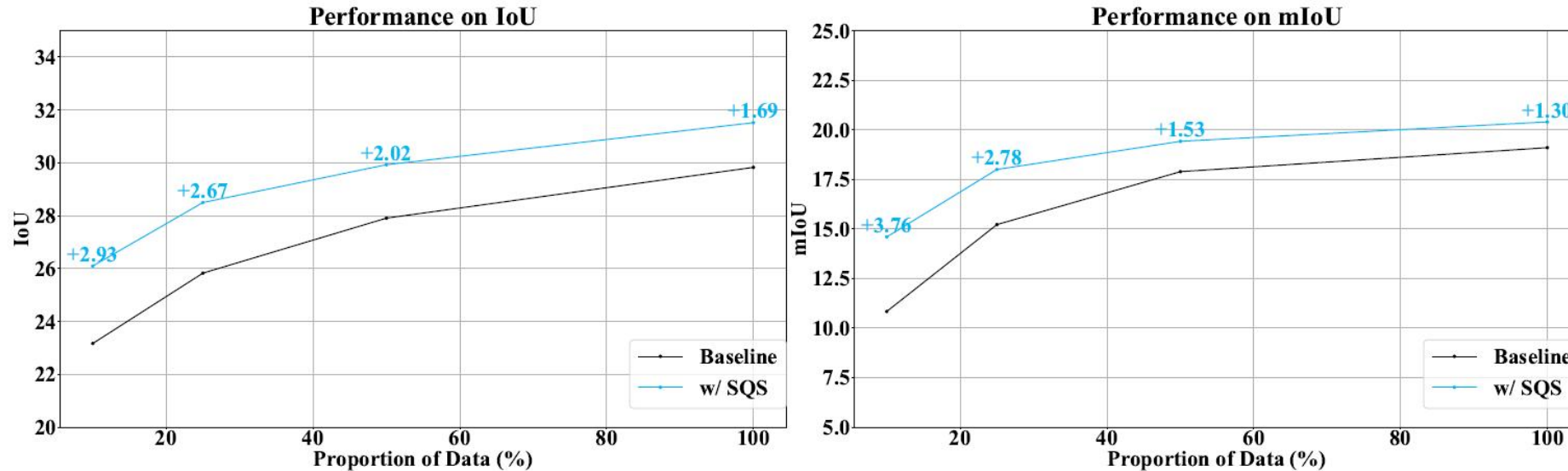
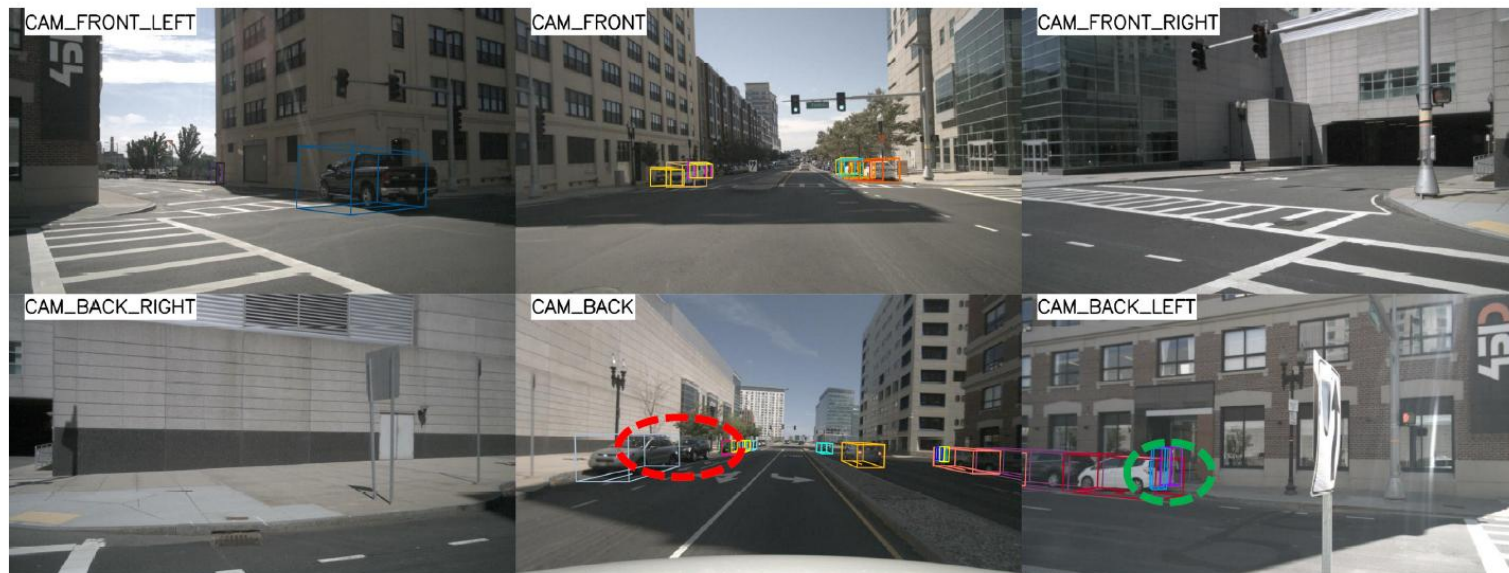


Figure 3: **Data efficiency analysis.** To assess data efficiency under limited annotation scenarios, we reduce the amount of labeled data used for downstream fine-tuning in the 3D semantic occupancy prediction task. The outcomes demonstrate that our pre-training method significantly enhances performance, even when only a small portion of annotations is available.

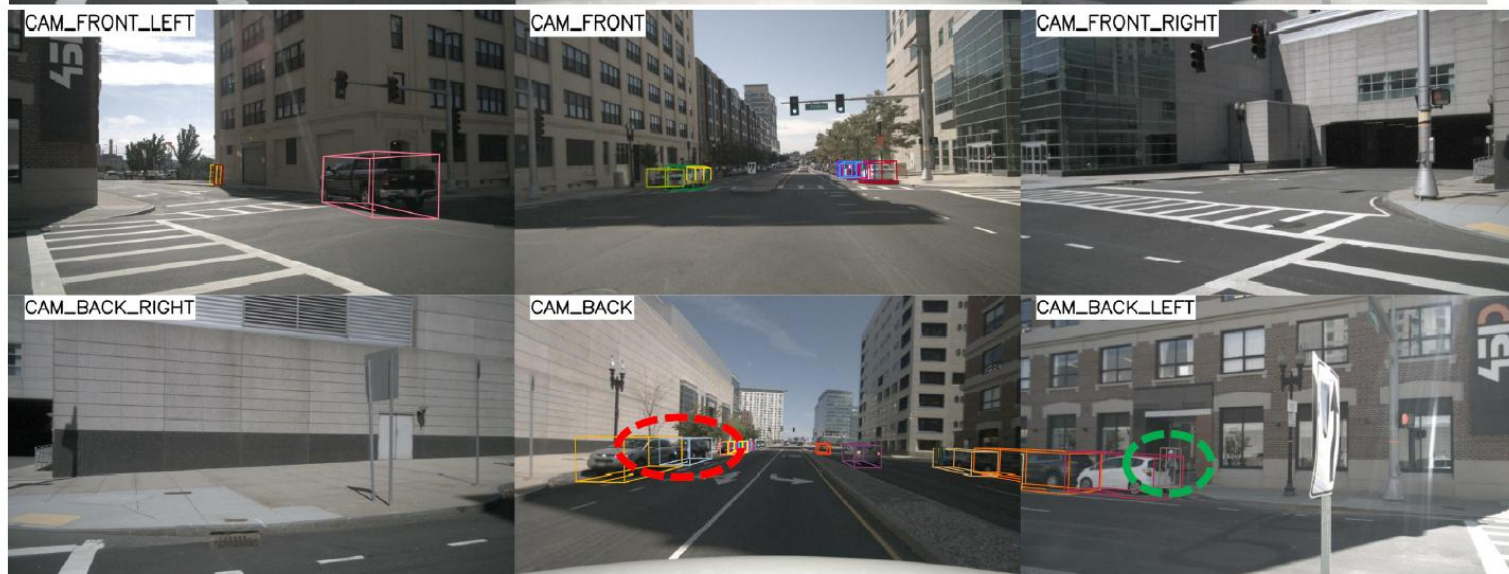
Visualization

3D Object Detection

SparseBEV

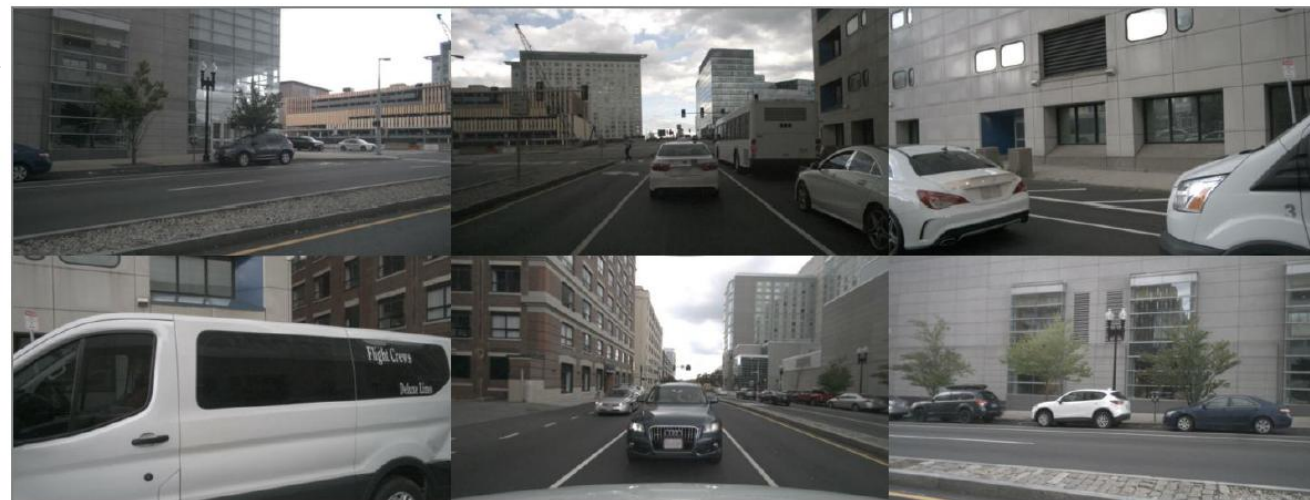


Ours

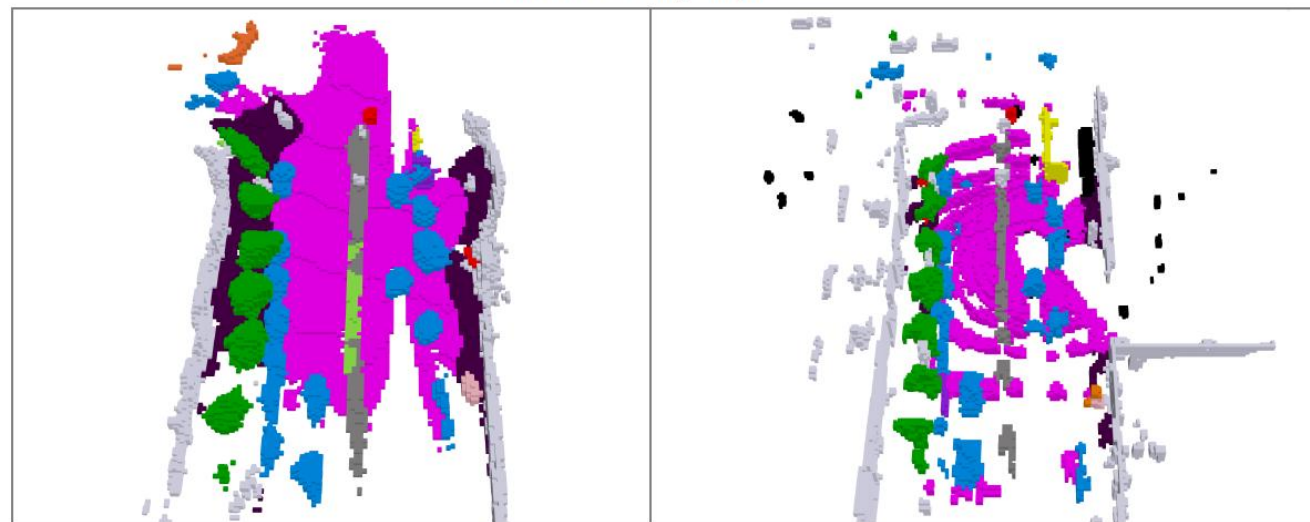


Visualization

3D Occupancy Prediction



Multi-view Image Inputs



Occupancy Prediction

Occupancy Ground Truth

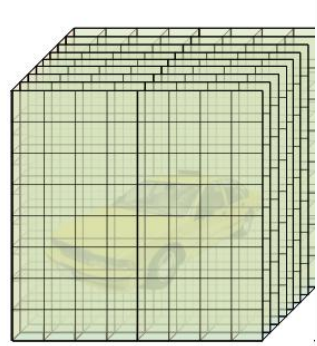


DLWM: Dual Latent World Models enable Holistic Gaussian-centric Pre-training in Autonomous Driving

Yiyao Zhu¹, Ying Xue², Haiming Zhang², Guangfeng Jiang³, Wending Zhou⁴, Xu Yan⁴, Jiantao Gao⁴, Yingjie Cai⁴, Bingbing Liu⁴,
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Scene representations for AD



Voxel-based

+ **Pros:** Detailed 3D geometry and precise spatial occupancy.

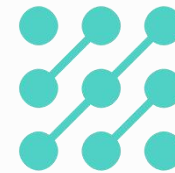
- **Cons:** Extremely high memory and computational overhead.



BEV-based

+ **Pros:** Computationally efficient 2D representation.

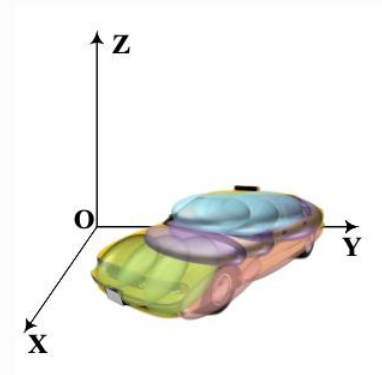
- **Cons:** Sacrifices vertical details and dense 3D geometry information.



Sparse-query

+ **Pros:** Very fast inference speeds with transformer-based architectures.

- **Cons:** Only provides coarse, object-centric scene knowledge.



Gaussian-centric

Set of 3D semantic Gaussians

Advantage: offers an optimal balance of detail and efficiency.

Problems for Gaussian-centric Models



Challenge 1: Scalability

Gaussian-centric methods rely on extensive manual annotations, which is expensive and limits **scalability** for large-scale real-world applications.



Challenge 2: Temporal Coherence

Existing pre-training focuses primarily on static 3D geometry, but fails to explicitly learn and model **temporal dynamics**.



Challenge 3: Latent Mismatch

There is no one-to-one correspondence between Gaussian queries in consecutive frames because of **permutation equivalence**.

Our Solution: DLWM (Dual Latent World Models)

A Novel Two-Stage Pre-training Paradigm

Goal

Enable holistic Gaussian-centric pre-training for all downstream tasks (perception, forecasting, planning).

Key Ideas

1. Apply self-supervised pre-training for **scalability**.
2. Integrate latent world model for **temporal dynamics**.
3. Employ BEV features for **latent supervision**.

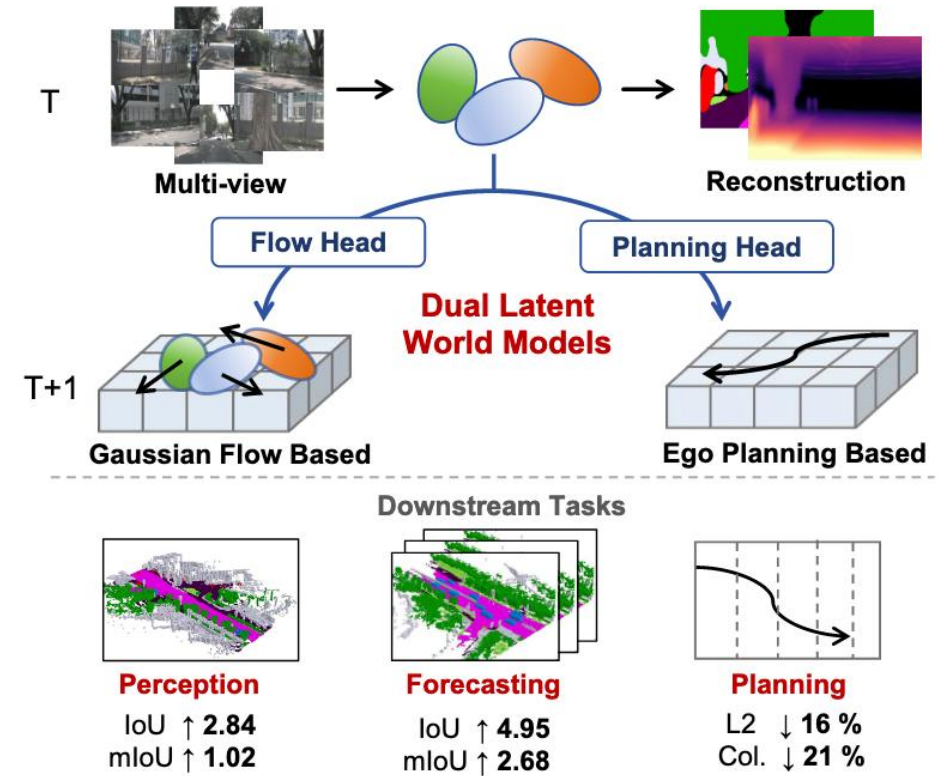


Figure 1. Illustration of our DLWM for pre-training and performance improvements for downstream tasks.

Method: Overall Pipeline (DLWM)

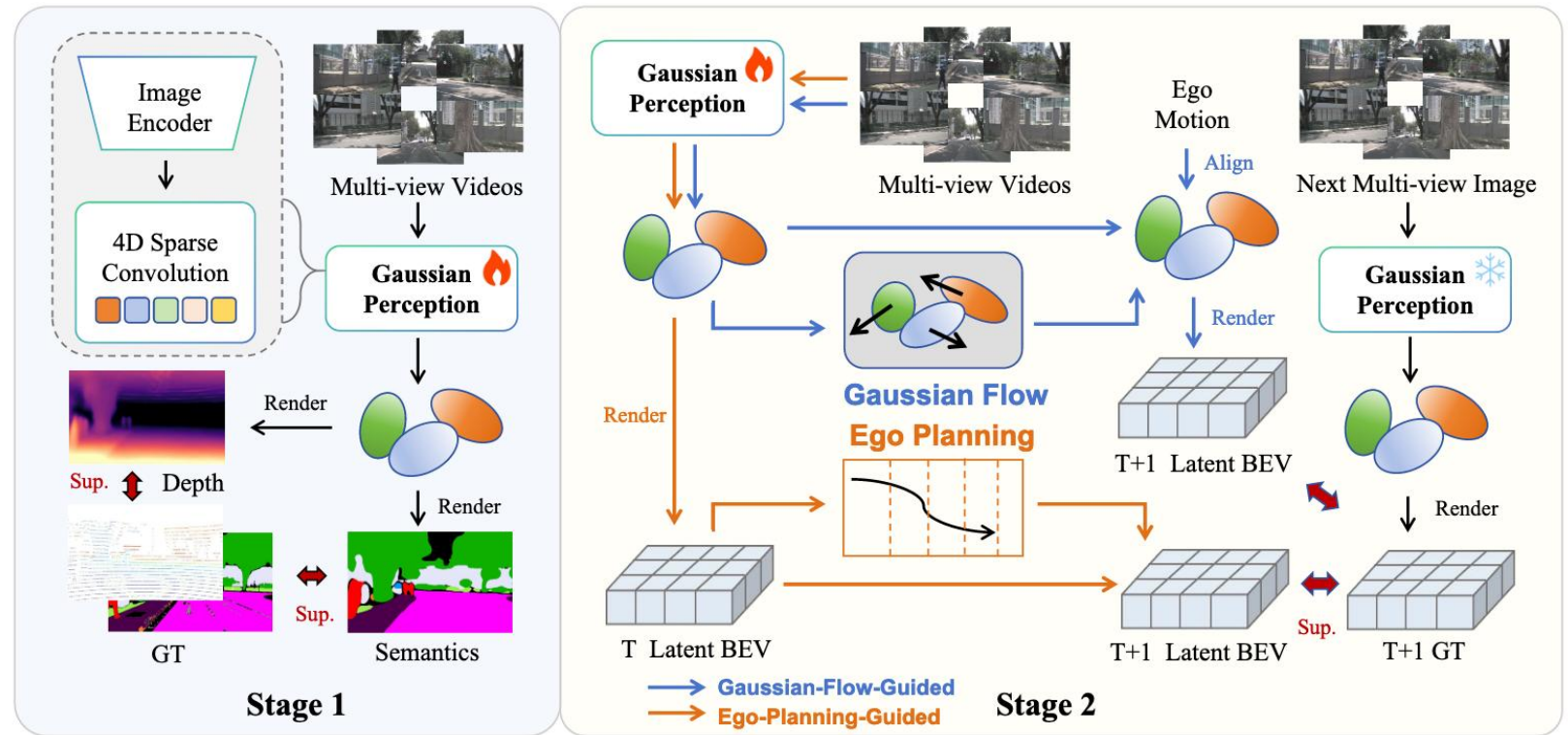


1. Learn Static Representation

Learn 3D Gaussian scene representations from multi-view videos using self-supervised reconstruction on depth and semantic maps.

2. Learn Dynamic Evolution

Train two parallel latent world models to learn the dynamic evolution patterns of the scene, thereby serving different downstream task requirements.



Core Idea: Two-stage Self-supervised Pre-training

DLWM-Stage 1: Gaussian Representation Learning

🕒 Objective

Learn fine-grained geometric and semantic features of 3D Gaussians to establish a foundational representation for modeling complex driving environments.

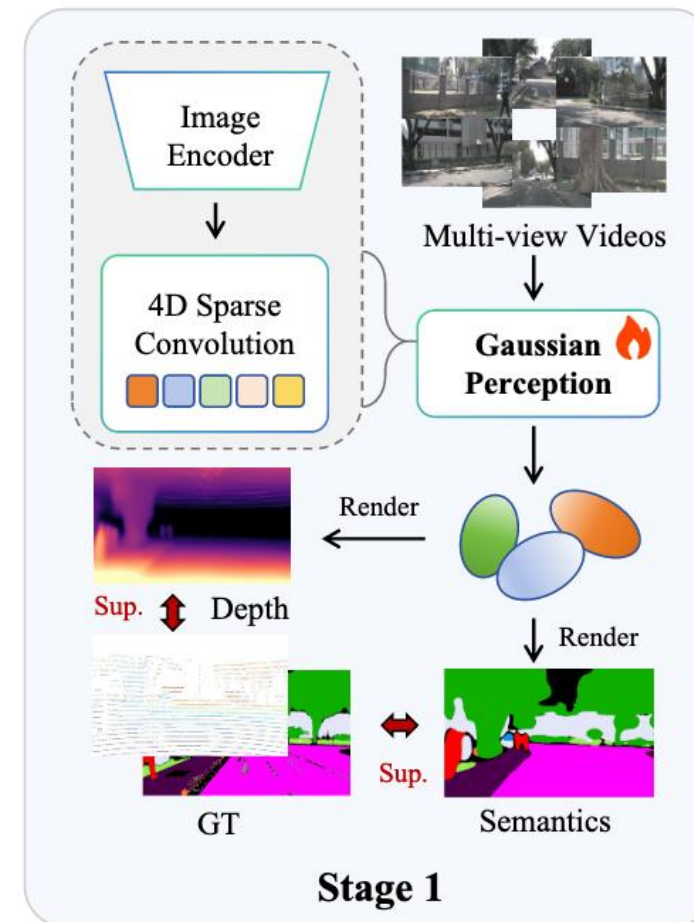
🔗 Method for Self-supervised Reconstruction

1. Input: Multi-view videos of driving scenes.
2. Perception Module: Predicts a set of 3D Gaussians (shape, position, attributes).
3. Rendering: Reconstructs input depth and semantic maps from the predicted Gaussians.

📈 Loss for Depth and Semantic Maps

$$\mathcal{L}_{rec} = \omega_1 \mathcal{L}_d + \omega_2 \mathcal{L}_{pd} + \omega_3 \mathcal{L}_{sem}$$

◆ **Key Advantage: No human annotations required.**



Learning by Self-supervised Reconstruction

DLWM-Stage 2: Dual Latent World Models (Model A)



Model A: Gaussian-Flow-Guided Latent World Model

Purpose: To learn and model the physical dynamics of the scene for downstream tasks, including **3D Occupancy Perception** and **4D Occupancy Forecasting** in autonomous driving scenarios.

Core Propagation & Supervision Process

1. Predict Flow

Generate displacement vectors for each 3D Gaussian from time t .

2. Propagate

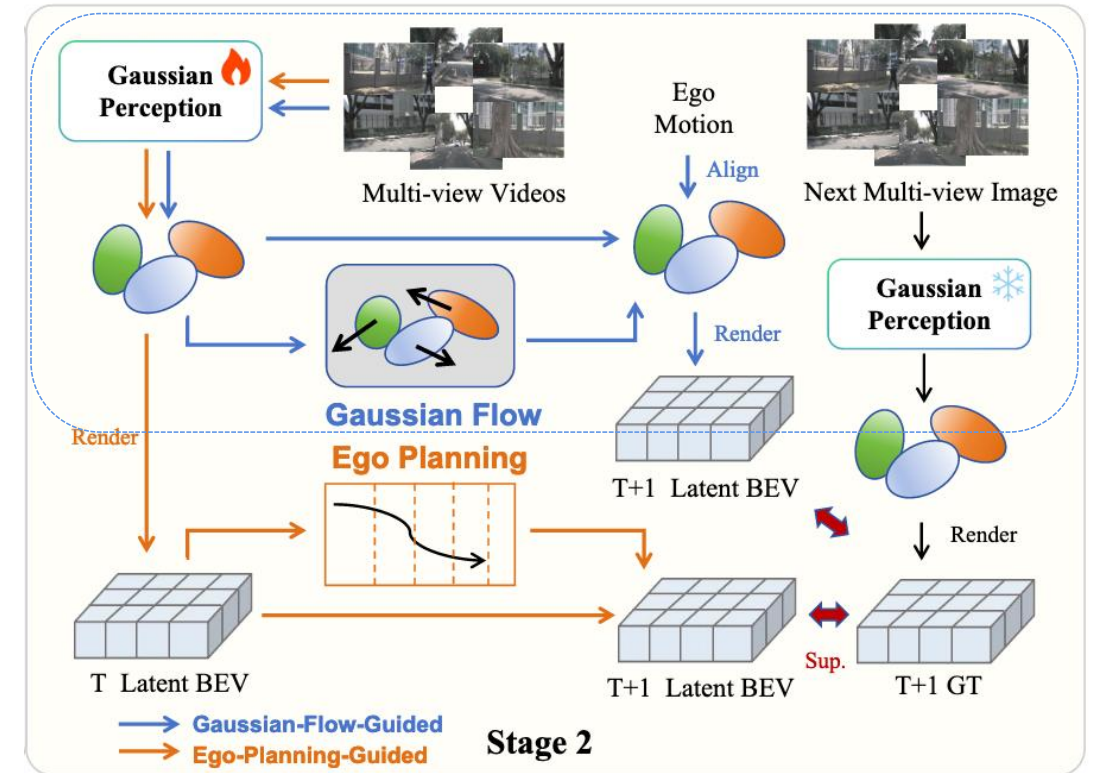
Project Gaussians to $t+1$ using the flow and ego-motion alignment.

3. Rasterize

Render propagated Gaussians to get the predicted future BEV latent ($\hat{B}_{t+1}$).

4. Supervise

Compare the prediction ($\hat{B}_{t+1}$) against the ground-truth BEV latent (B_{t+1}).



Pipeline of Gaussian-Flow-Guided World Model

DLWM-Stage 2: Dual Latent World Models (Model A)

Model A: Downstream Tasks

3D Occupancy Perception:

1. Obtain current 3D Gaussians from pre-trained perception module.
2. Gaussian-to-Occupancy Splatting for occupancy prediction.

4D Occupancy Forecasting:

1. Align current 3D Gaussians to the next frame based on the ego trajectory.
2. The newly-observed areas are completed with random Gaussians.
3. Employ 3D sparse convolution followed by refinement layers to model the current Gaussians and the newly-completed Gaussians.

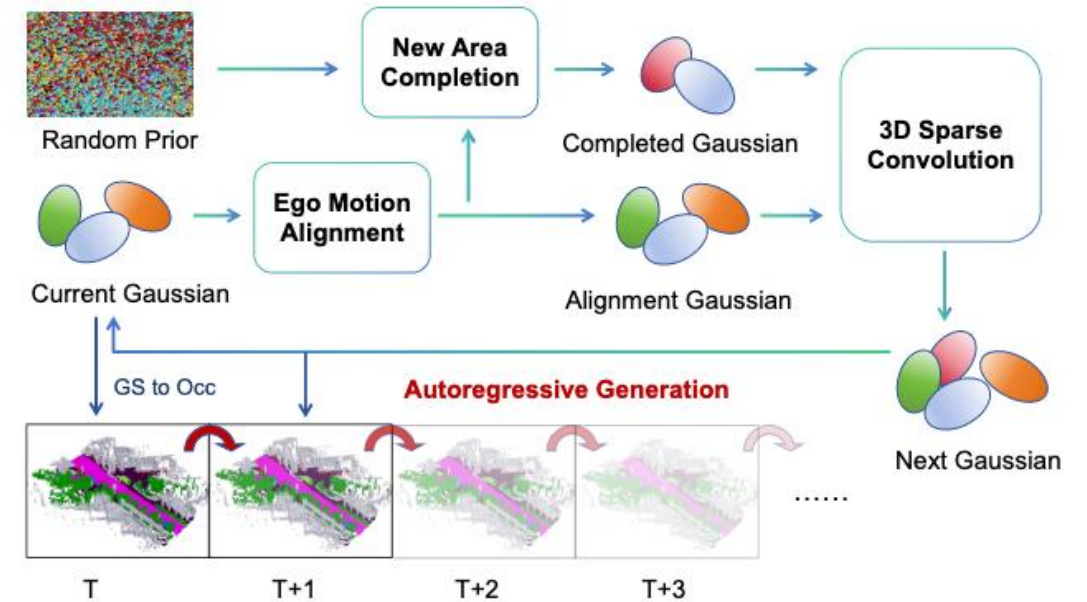


Figure 3. **4D occupancy forecasting in a stream manner.** We transform current 3D Gaussians to the next frame with ego motion alignment and complete new area with random Gaussians.

DLWM-Stage 2: Dual Latent World Models (Model B)

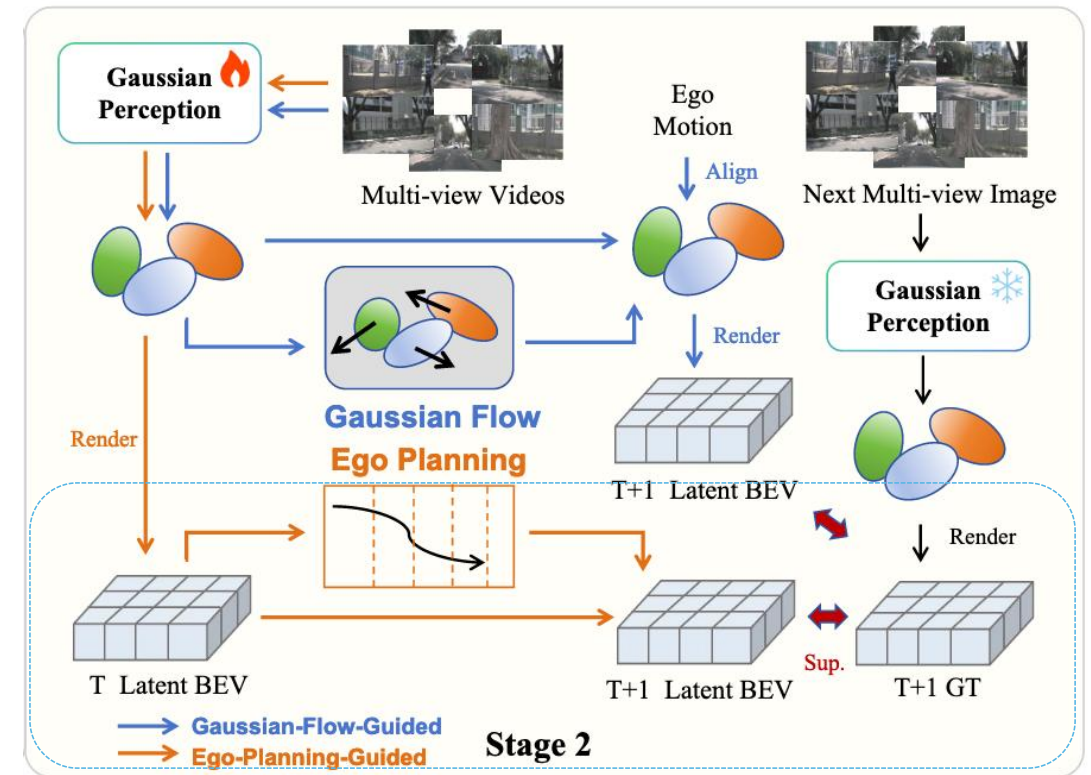
Model B: Ego-Planning-Guided Latent World Model

🎯 **Purpose:** Optimized for **Motion Planning** task, ensuring future scene predictions are actionable.

📋 **Process:**

1. Predict ego trajectory ($\mathbf{T}$) using current Gaussian scene features.
2. Condition future BEV latent prediction ($\mathbf{B}_{t+1}$) on this trajectory.
3. Supervise prediction against ground-truth BEV ($\mathbf{B}_{t+1}$).

💡 **Key Insight:** Creates a **motion-aware context** where planning objectives guide temporal scene learning.



Pipeline of Ego-Planning-Guided World Model

Experimental Setup



1 / Datasets

nuScenes: A large-scale driving dataset with 1000 meticulously annotated sequences covering diverse driving scenarios.

SurroundOcc: Extended nuScenes with dense 3D semantic occupancy annotations across 18 classes.



2 / Evaluation Metrics

Perception & Forecasting:

- **mIoU / IoU:** Mean Intersection over Union for overall and class-wise 3D space understanding.

Planning:

- **L2 Error:** Measures trajectory prediction accuracy.
- **Collision Rate:** Assesses the safety of generated paths.



3 / Implementation

Backbone Network:

ResNet101 image encoder followed by Feature Pyramid Network (FPN).

Number of Gaussians:

25,600 (for high-fidelity representation).

Optimizer:

AdamW with learning rate warming up.

Main Results – 3D Occupancy Perception

Baseline (w/o pre-training)

20.83 mIoU

DLWM (Ours)

21.85 mIoU

+1.02 mIoU improvement

Table 1. 3D occupancy perception results on the SurroundOcc-nuScenes validation set [38]. *: We re-evaluate the checkpoints released by GaussianWorld because they repeatedly calculated the metrics for intermediate time-interval frames within each video.

Method	IoU	mIoU	barrier	bicycle	bus	car	const. veh.	motorcycle	pedestrian	traffic cone	trailer	truck	drive. suf.	other flat	sidewalk	terrain	manmade	vegetation
MonoScene [3]	23.96	7.31	4.03	0.35	8.00	8.04	2.90	0.28	1.16	0.67	4.01	4.35	27.72	5.20	15.13	11.29	9.03	14.86
Atlas [31]	28.66	15.00	10.64	5.68	19.66	24.94	8.90	8.84	6.47	3.28	10.42	16.21	34.86	15.46	21.89	20.95	11.21	20.54
BEVFormer [23]	30.50	16.75	14.22	6.58	23.46	28.28	8.66	10.77	6.64	4.05	11.20	17.78	37.28	18.00	22.88	22.17	13.80	22.21
TPVFormer [11]	30.86	17.10	15.96	5.31	23.86	27.32	9.79	8.74	7.09	5.20	10.97	19.22	38.87	21.25	24.26	23.15	11.73	20.81
OccFormer [51]	31.39	19.03	18.65	10.41	23.92	30.29	10.31	14.19	13.59	10.13	12.49	20.77	38.78	19.79	24.19	22.21	13.48	21.35
SurroundOcc [39]	31.49	20.30	20.59	11.68	28.06	30.86	10.70	15.14	14.09	12.06	14.38	22.26	37.29	23.70	24.49	22.77	14.89	21.86
GaussianFormer [14]	29.83	19.10	19.52	11.26	26.11	29.78	10.47	13.83	12.58	8.67	12.74	21.57	39.63	23.28	24.46	22.99	9.59	19.12
GaussianFormer-2 [12]	31.74	20.82	21.39	13.44	28.49	30.82	10.92	15.84	13.55	10.53	14.04	22.92	40.61	24.36	26.08	24.27	13.83	21.98
GaussianWorld* [60]	32.77	21.79	21.61	13.30	27.28	31.21	13.89	16.91	13.28	11.77	14.82	23.66	41.91	24.31	28.35	26.32	15.67	24.54
Baseline	31.77	20.83	21.17	13.82	26.74	31.95	10.90	16.03	13.94	10.87	13.44	22.52	40.79	24.28	26.06	24.13	14.05	22.55
DLWM (ours)	34.61	21.85	21.71	13.56	27.65	31.22	12.50	16.93	12.35	11.42	13.96	24.12	42.27	24.94	27.39	26.36	16.61	26.64

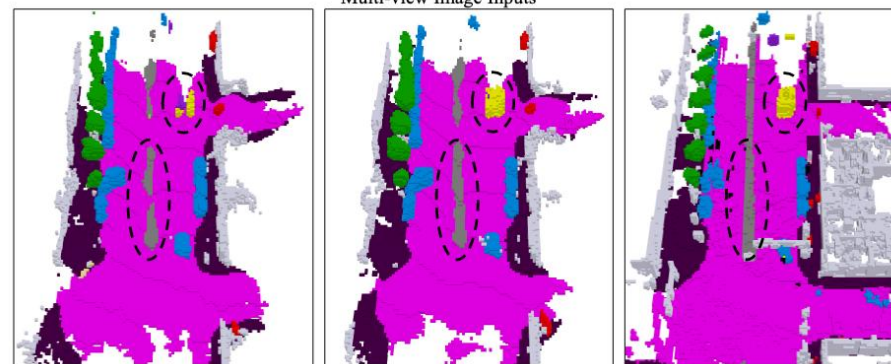


SOTA Comparison

Outperforms all previous methods, including Gaussian-based approaches on the SurroundOcc-nuScenes val set.



Multi-view Image Inputs



Occupancy Prediction w/o Pre-training Occupancy Prediction w/ Pre-training Occupancy GT

■ pedestrian ■ traffic cone ■ trailer ■ truck ■ barrier ■ bicycle ■ bus ■ drivable surface
■ motorcycle ■ construction vehicle ■ sidewalk ■ terrain ■ manmade ■ vegetation ■ flat ■ car

Figure 2. Qualitative comparison of 3D Occupancy Perception over GT, baseline without pre-training and ours with pre-training.

Main Results – 4D Occupancy Forecasting



Task Objective

Predict the evolution of the 3D scene up to **3 seconds** into the future on the SurroundOcc-nuScenes validation set.

Baseline (w/o pre-training)

15.09 Avg. mIoU

DLWM (Ours)

17.77 Avg. mIoU

+2.68 mIoU improvement



Establishes a New SOTA

The results validate the effectiveness of the Gaussian-flow-guided world model for modeling complex temporal dynamics in autonomous driving.

Table 2. **4D occupancy forecasting results on the SurroundOcc-nuScenes validation set [39]**. Aux. Sup. represents auxiliary supervision. Avg. computes the average result of 1s, 2s, and 3s.

Method	Input	Aux. Sup.	mIoU (%) ↑					IoU (%) ↑				
			0s	1s	2s	3s	Avg.	0s	1s	2s	3s	Avg.
Copy&Paste	3D-Occ	None	66.38	14.91	10.54	8.52	11.33	62.29	24.47	19.77	17.31	20.52
OccWorld-O [51]	3D-Occ	None	66.38	25.78	15.14	10.51	17.14	62.29	34.63	25.07	20.18	26.63
OccWorld-T [51]	Camera	Semantic LiDAR	7.21	4.68	3.36	2.63	3.56	10.66	9.32	8.23	7.47	8.34
OccWorld-S [51]	Camera	None	0.27	0.28	0.26	0.24	0.26	4.32	5.05	5.01	4.95	5.00
OccWorld-D [51]	Camera	3D-Occ	18.63	11.55	8.10	6.22	8.62	22.88	18.90	16.26	14.43	16.53
PreWorld [22]	Camera	2D Labels & 3D-Occ	-	12.27	9.24	7.15	9.55	-	23.62	21.62	19.63	21.62
Baseline	Camera	3D-Occ	21.12	18.18	15.11	11.97	15.09	32.12	29.98	25.43	21.55	25.65
DLWM (ours)	Camera	3D-Occ	22.21	19.66	17.79	15.87	17.77	33.47	32.63	30.68	28.49	30.60

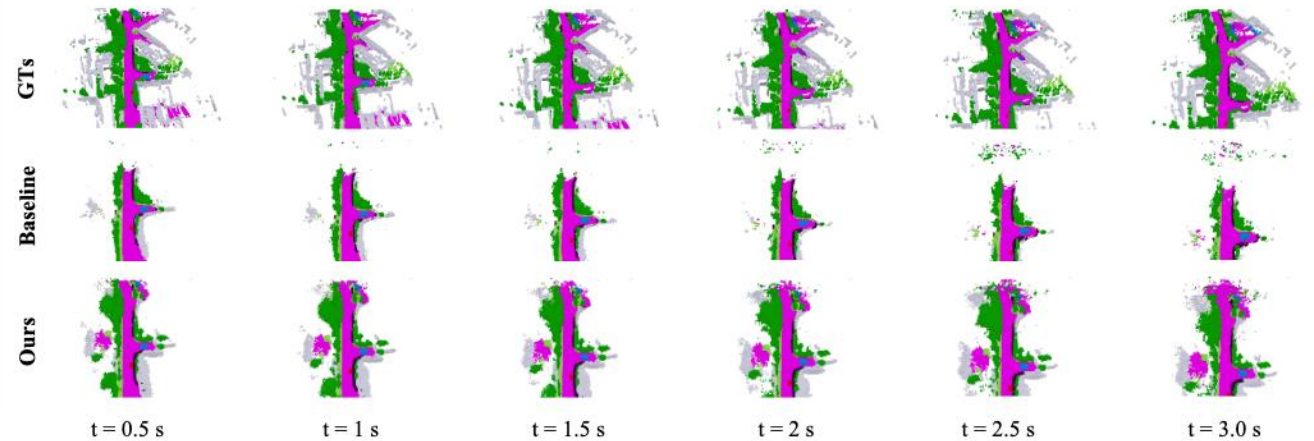


Figure 3. **Qualitative comparison of 4D Occupancy Forecasting over GTs, baseline without pre-training and ours with pre-training.**

Main Results – Motion Planning



Task Objective

Plan a safe and accurate ego trajectory over a **3-second horizon** on the nuScenes validation set.



Baseline Model

Without Pre-training

0.55 m

Average L2 Error



DLWM (Ours)

0.46 m

↓ **16% Improvement**



SOTA Performance & Conclusion

DLWM achieves a **state-of-the-art L2 error of 0.46m**, demonstrating competitive performance with methods that utilize multiple auxiliary tasks, proving the efficacy of the pre-training with a planning-guided world model.

Table 3. **Motion Planning Results on the nuScenes validation set.** The metrics are computed by the way in VAD [16]. $\diamond$: Lidar-based methods. We do not utilize ego status in the planning module.

Method	Aux. Task	L2 (m) ↓				Collision Rate (%) ↓			
		1s	2s	3s	Avg.	1s	2s	3s	Avg.
NMP $\diamond$ [47]	Det & Motion	0.53	1.25	2.67	1.48	0.04	0.12	0.87	0.34
FF $\diamond$ [7]	FreeSpace	0.55	1.20	2.54	1.43	0.06	0.17	1.07	0.43
EO $\diamond$ [20]	FreeSpace	0.67	1.36	2.78	1.60	0.04	0.09	0.88	0.33
ST-P3 [8]	Det & Map & Depth	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71
UniAD [10]	Det&Track&Map&Motion&Occ	0.44	0.67	0.96	0.69	0.04	0.08	0.23	0.12
VAD-Tiny [16]	Det & Map & Motion	0.46	0.76	1.12	0.78	0.21	0.35	0.58	0.38
VAD-Base [16]	Det & Map & Motion	0.41	0.70	1.05	0.72	0.07	0.17	0.41	0.22
PARA-Drive [40]	Det&Track&Map&Motion&Occ	0.25	0.46	0.74	0.48	0.14	0.23	0.39	0.25
GenAD [52]	Det & Map & Motion	0.28	0.49	0.78	0.52	0.08	0.14	0.34	0.19
OccWorld [51]	3D-Occ	0.39	0.73	1.18	0.77	0.11	0.19	0.67	0.32
BEV-Planner [25]	None	0.28	0.42	0.68	0.46	0.04	0.37	1.07	0.49
LAW [23]	None	0.26	0.57	1.01	0.61	0.14	0.21	0.54	0.30
Baseline	None	0.27	0.51	0.87	0.55	0.11	0.17	0.46	0.24
DLWM (ours)	None	0.21	0.43	0.76	0.46	0.11	0.14	0.32	0.19

Conclusion



We proposed DLWM

A novel two-stage self-supervised pre-training paradigm for Gaussian-centric representation in autonomous driving, aiming to learn unified 3D scene understanding.



DLWM unifies spatial & temporal learning

Integrates spatial understanding (Stage 1) and motion reasoning (Stage 2) through dual latent world models, bridging static scene and dynamic behavior modeling.



Achieves State-of-the-Art Results

Significantly outperforms existing methods on downstream 3D occupancy perception, 4D occupancy forecasting, and motion planning tasks.



Key Contribution

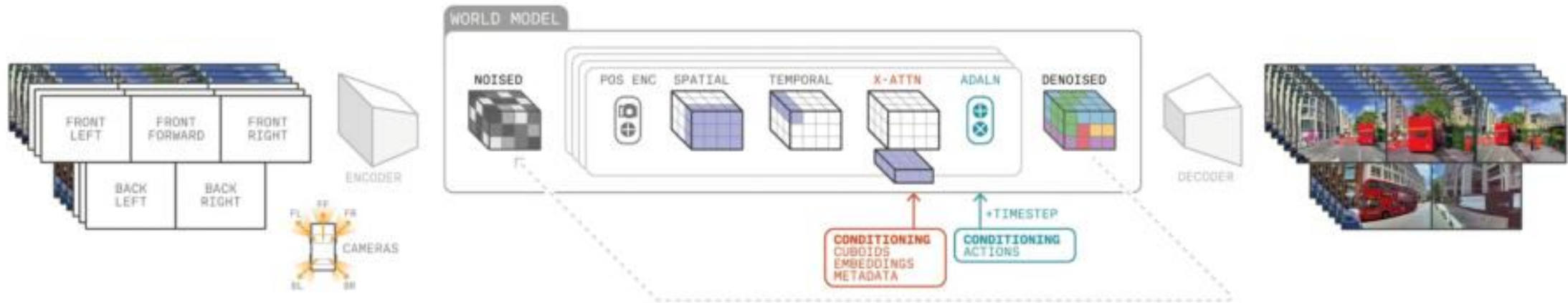
The essential and scalable value of holistic Gaussian-centric pre-training is building more capable and robust autonomous driving systems.

Outline

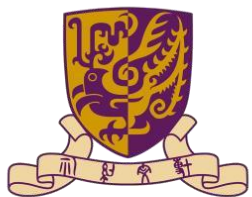
Robust and Generalized 3D Understanding for L3/L4 AD and Embodied AI

- ◆ Multimodal 3D Data
 - Paired 3D-Text Native Data and Benchmark
- ◆ Pre-training
 - Vision Pre-training for autonomous Driving
- ◆ Robust and Generalized 3D Perception for Autonomy System
 - Robust Perception for OOD Scenarios

World Model GAIA-2



**Wayve GAIA-2 with 2.6B
dollar financing**
<https://wayve.ai/thinking/gaia-2/>



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



未来智联网络研究院



InfiniVerse: Occupancy Guided Unbounded Scene Generation for Autonomous Driving

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Guanyi Zhao⁴, Ying-Cong Chen³, Bingbing Liu⁴, Shuguang Cui^{2,1}, Zhen Li^{2,1}†✉*

¹ The Future Network of Intelligence Institute, The Chinese University of Hong Kong (Shenzhen),

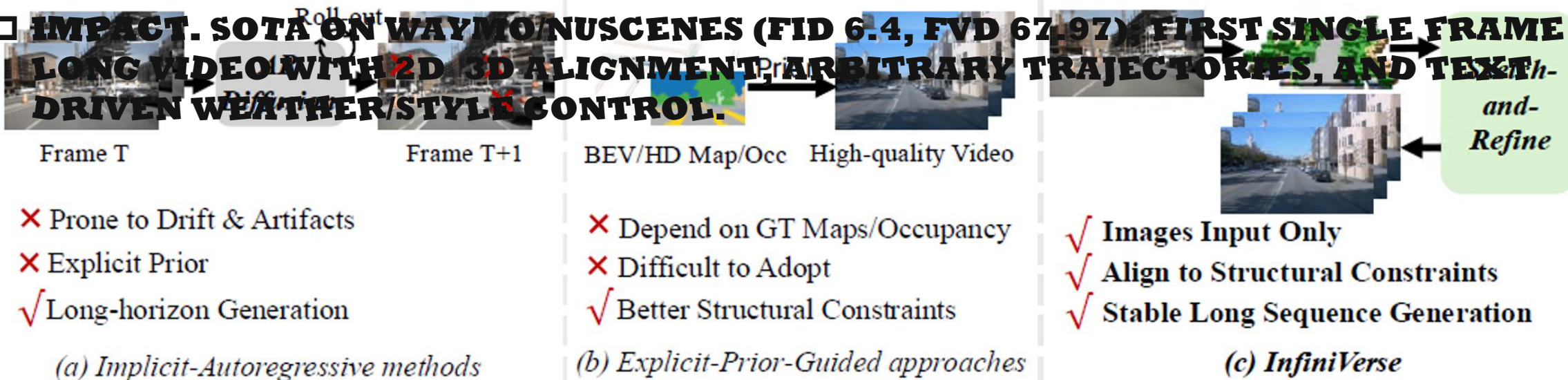
² School of Science and Engineering, The Chinese University of Hong Kong (Shenzhen),

³ HKUST,

⁴ Huawei Noah's Ark Lab

Challenges

- ❑ **CHALLENGE. EXISTING METHODS LACK LONG HORIZON CONSISTENCY, GEOMETRIC STABILITY, AND CONTROLLABILITY EITHER ACCUMULATING ERROR WITHOUT EXPLICIT 3D, OR RELYING ON BEV/HD MAPS THAT STRUGGLE WITH LONG VIDEO.**
- ❑ **APPROACH. INFINIVERSE REBUILDS 3D OCCUPANCY FROM A SINGLE MULTI VIEW FRAME AND EXPANDS ALONG ARBITRARY TRAJECTORIES VIA SKETCH AND REFINE: COARSE OCCUPANCY GUIDES VIDEO, THEN VIDEO REFINES 3D CLOSING THE 2D 3D LOOP.**
- ❑ **IMPACT. SOTA ON WAYMO/NUSCENES (FID 6.4, FVD 67.97), FIRST SINGLE FRAME LONG VIDEO WITH 2D 3D ALIGNMENT, ARBITRARY TRAJECTORIES, AND TEXT-DRIVEN WEATHER/STYLE CONTROL.**



Method

- Inputs: Single multi-view frame
- Outputs: Long-horizon video and detailed 3D occupancy
- Application: AD Simulator, Low cost data augmentation

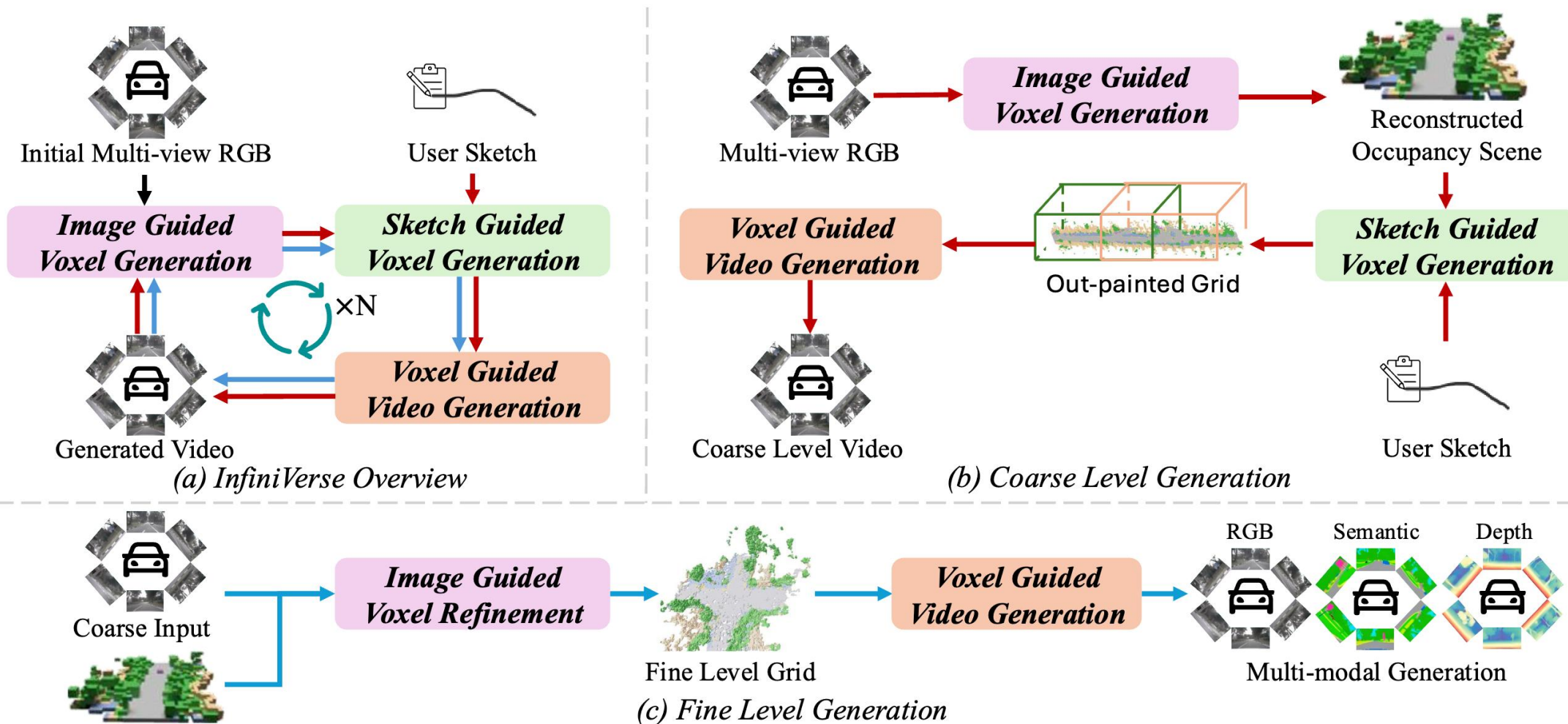




Figure 3. Demonstration on video quality on nuScenes.

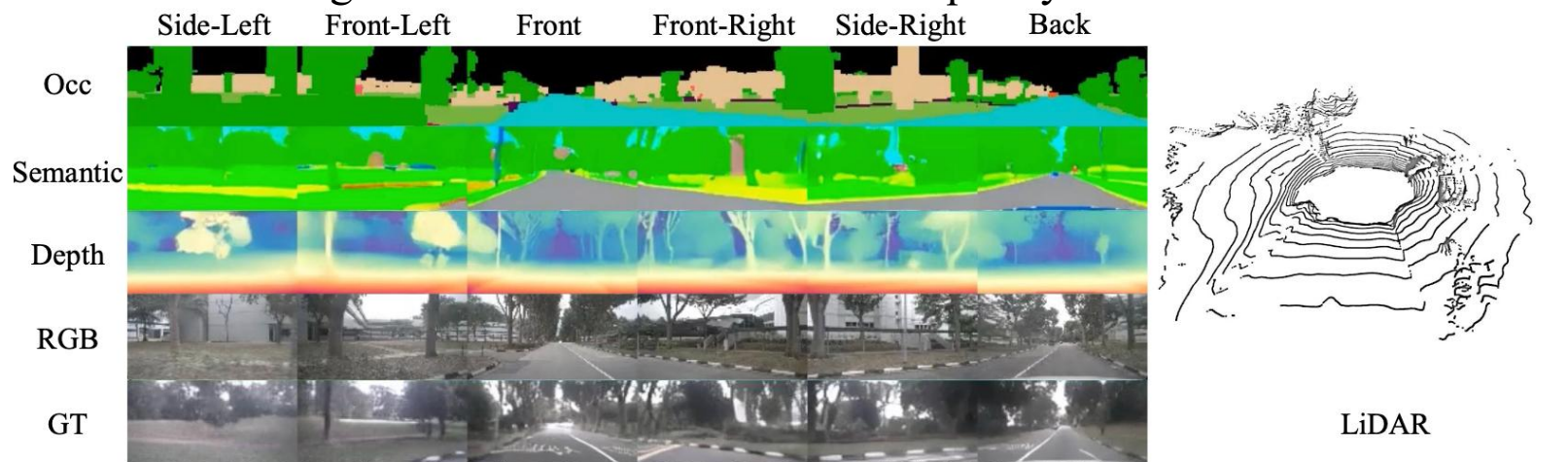


Figure 4. Demonstration on aligned multiple multi-view sensor output compared with GT RGB Frame on nuScenes.

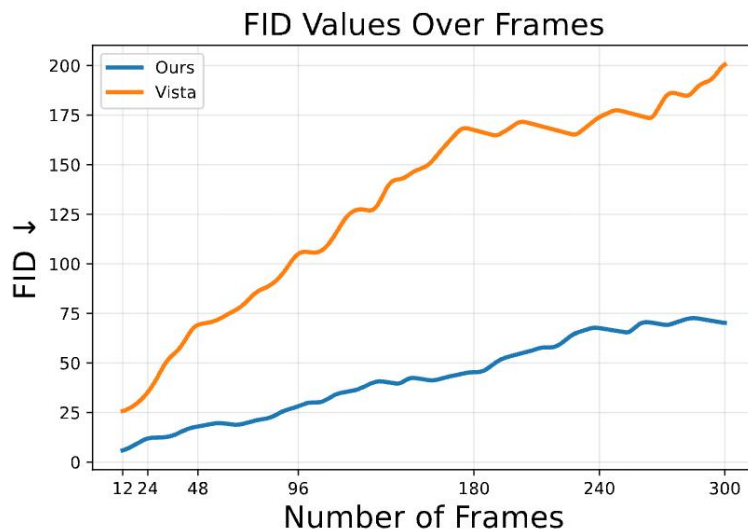


Results

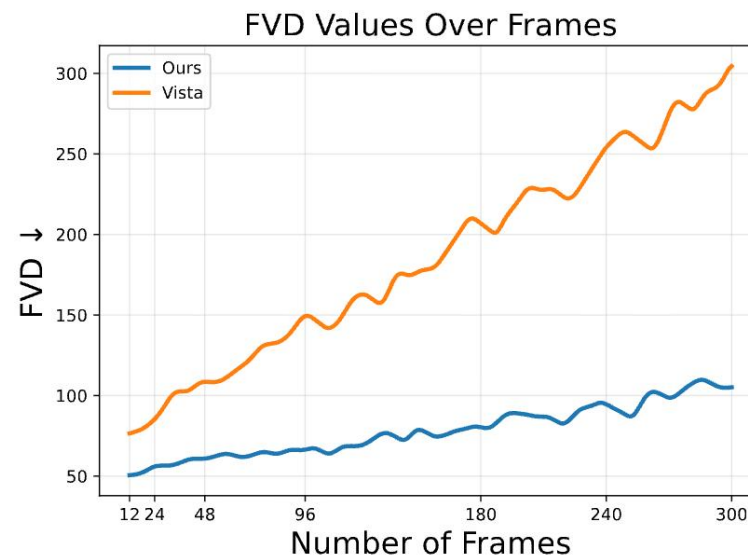
Method	Input Type	Output Type		2D-3D Aligned	FID↓
		MV Video	3D Repre.		
BEVGen [38]	BEV Map	×	×	-	40.48
BEVControl [50]	BEV Sketch	×	×	-	29.04
GenAD [49]	Image+BEV map	×	×	-	23.90
Drive-WM [44]	Image+Action+Box/Map	✓	×	-	15.8
DriveDreamer-2 [55]	Box+FoV Map	✓	HD Map+BEV	×	25.0
MagicDrive [11]	Pose+Box+BEV	✓	×	-	16.59
MagicDrive-V2 [10]	Traj.+Box+BEV	✓	×	-	20.91
MagicDrive-3D [9]	Traj.+Box+BEV	✓	3DGS	×	20.67
InfiniCity [21]	Voxel	×	Voxel	×	77.0
Panacea [45]	Image+Box+FoV Map	✓	×	-	16.87
Vista [12]	Action	×	×	-	13.97
Uniscene [20]	BEV	✓	Voxel+LiDAR	×	6.45
DiVE [16]	Traj.+Box+FoV	✓	×	-	10.68
DelPhi [26]	BEV Map	✓	×	-	15.08
X-Scene [51]	Text+Box+HD Map	✓	Voxel	×	12.77
InfiniCube [25]	Box+HD Map	×	Voxel+3DGS	×	80.13
DriveGan [18]	Image+Action	×	×	-	73.4
WovoGen [24]	Image+HD Map+Voxel	✓	HD Map+Voxel	×	27.6
DriveDreamer [43]	Image+Action+FoV Map	✓	×	-	14.9
InfiniVerse (ours)	Image	✓	Voxel+LiDAR	✓	6.40

Table 1. Comparison of input/output between previous method and ours.

Results



(a) FID values over frames on Waymo.



(b) FVD values over frames on nuScenes.

Table 2. Long-horizon generation quality under increasing rollout horizons. InfiniVerse shows a substantially slower degradation trend than Vista in both frame-level fidelity and temporal consistency.

Results

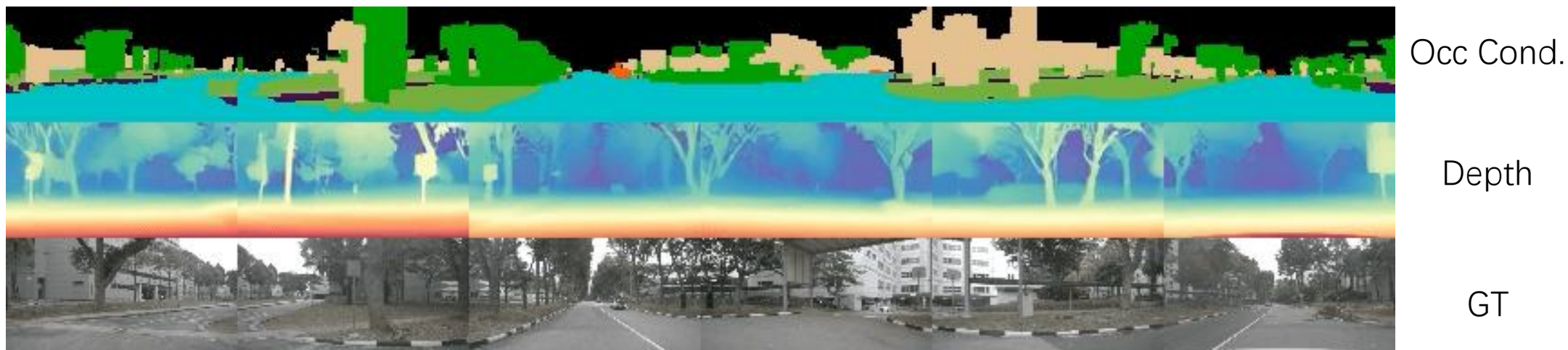


Figure 5. Demonstration on generated depth on nuScenes.

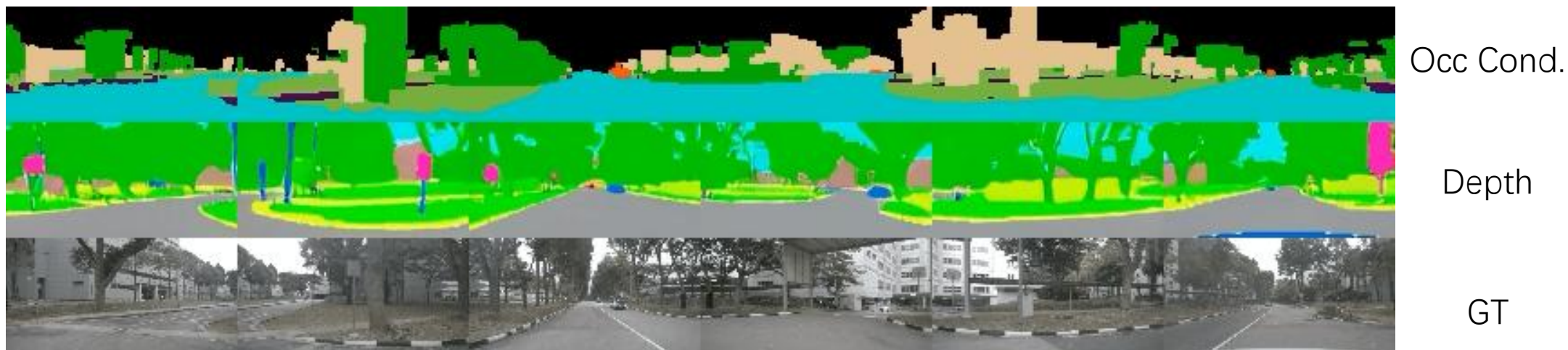
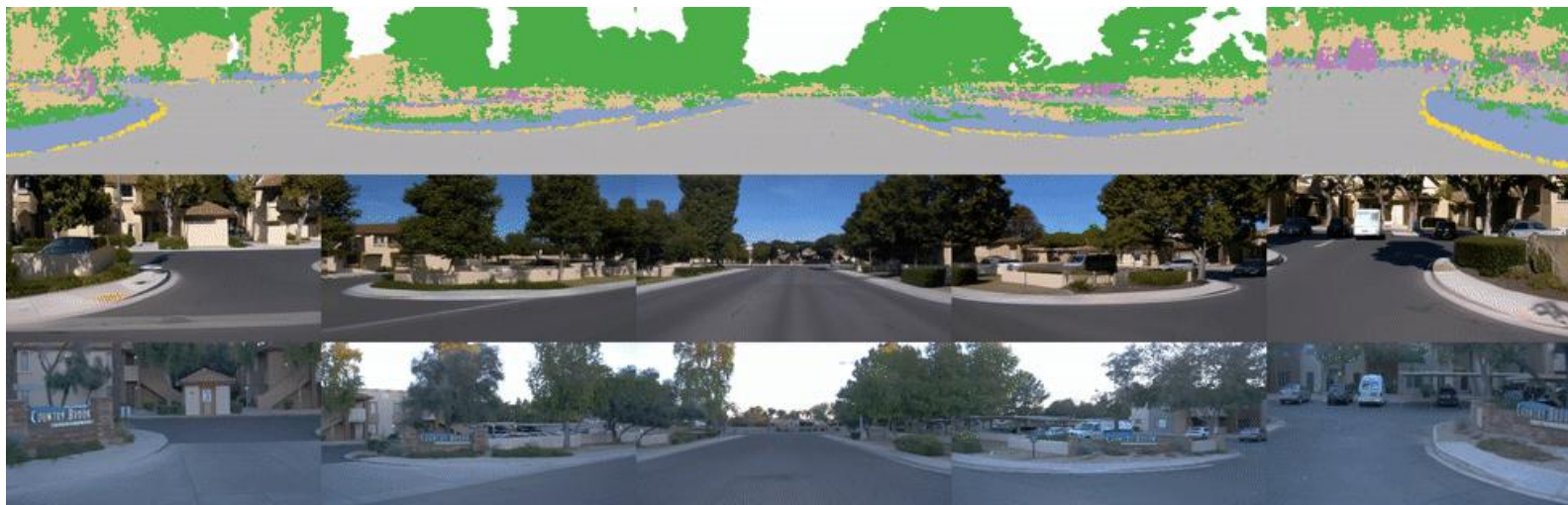


Figure 6. Demonstration on generated semantics on nuScenes.

Results

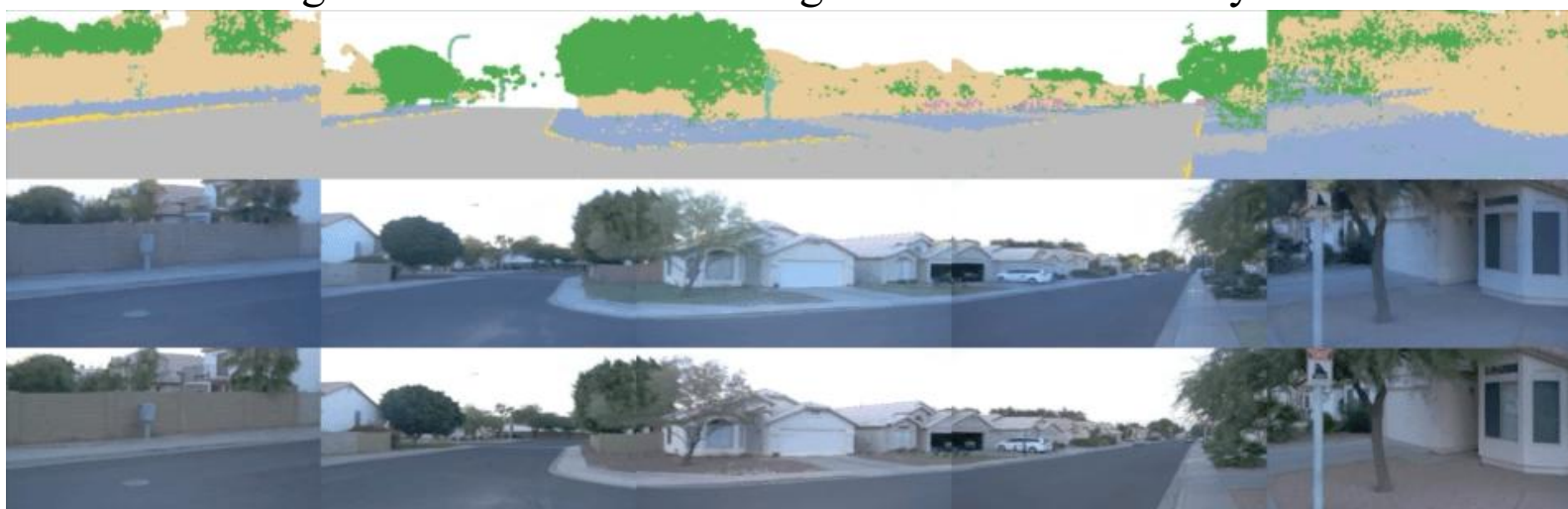


Occ Cond.

Depth

GT

Figure 7. Demonstration on generated video on waymo.

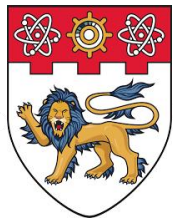


Occ Cond.

Depth

GT

Figure 8. Demonstration on Long horizon video on Waymo.



DriveGEN: Generalized and Robust 3D Detection in Driving via Controllable Text-to-Image Diffusion Generation

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1 FNii-Shenzhen 2 SSE, CUHK-Shenzhen 3 National University of Singapore

4 Nanyang Technological University 5 Baoji University of Arts and Sciences 6 Sun Yat-sen University

Challenges

- High data collection costs and diverse real-world scenarios **limit the variety of training data**, especially for the tailed cases (e.g., various weather conditions)
- Once distribution shifts exist, well-trained model suffers substantial performance degradation
- Significant performance degradation on OOD test data may result in traffic accidents and serious safety risks

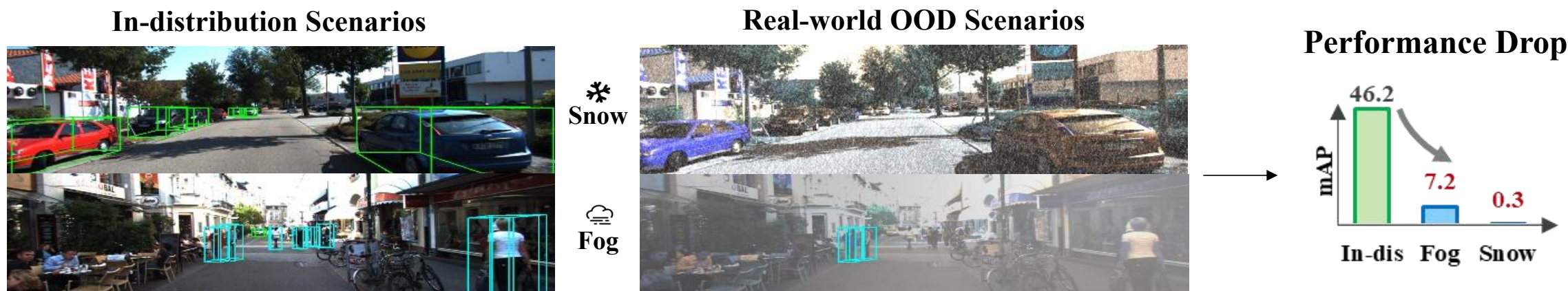


Figure 2. An illustration of the model performance degradation on out-of-distribution (OOD) data

Motivation

Could we devise a training-free controllable Text-to-Image (T2I) diffusion generation method for training data augmentation, thereby improving the robustness of 3D detectors?

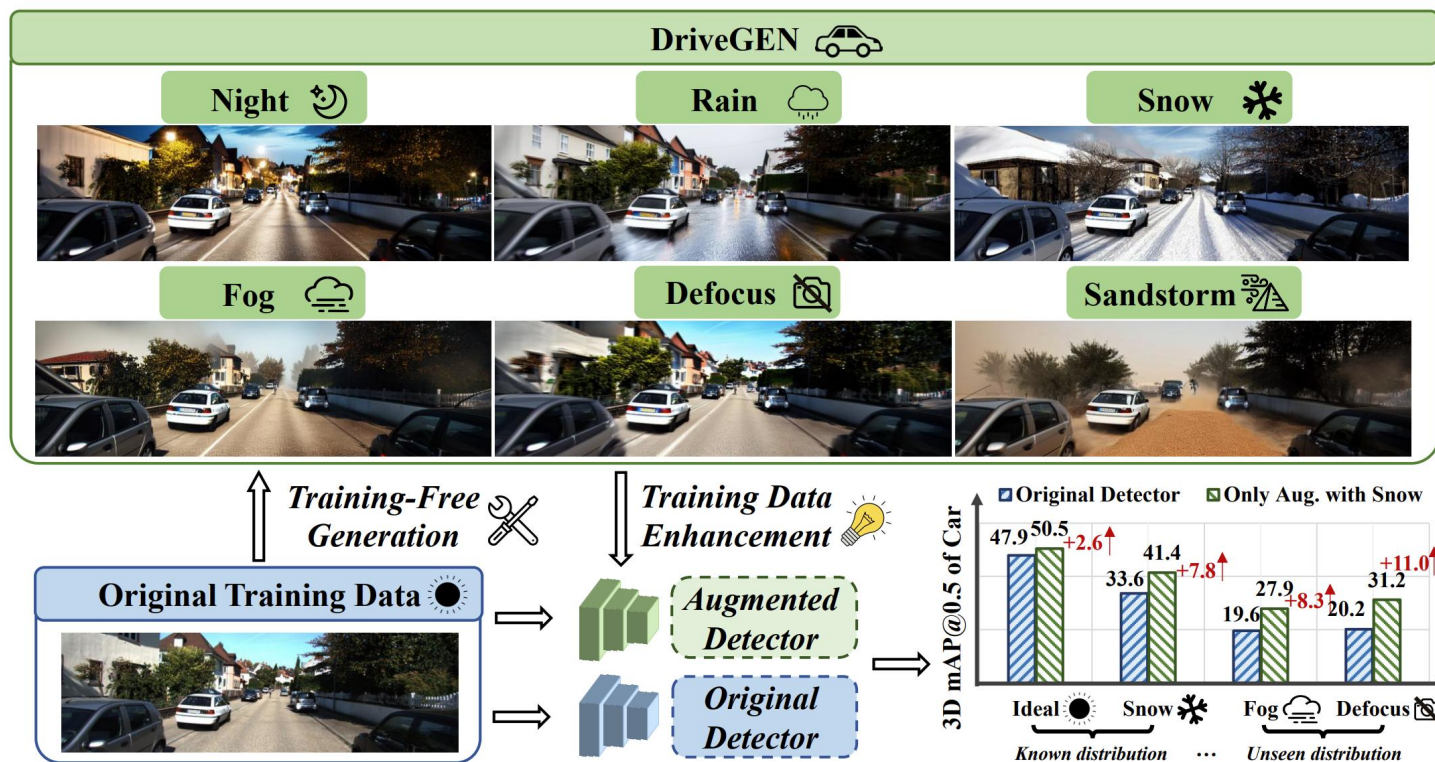
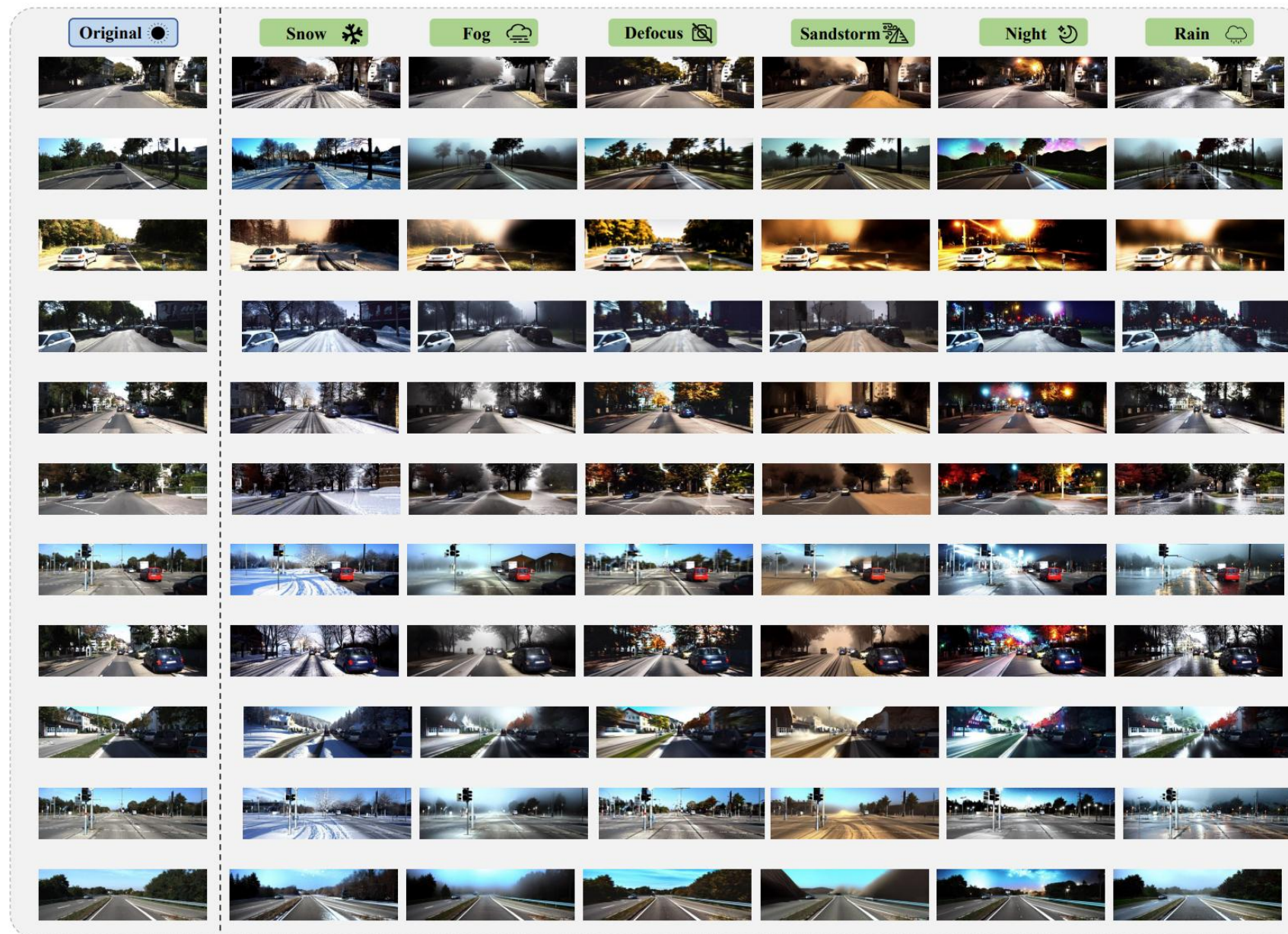


Figure 3. An illustration of DriveGEN for enhancing vision-centric 3D detectors.



Visualization

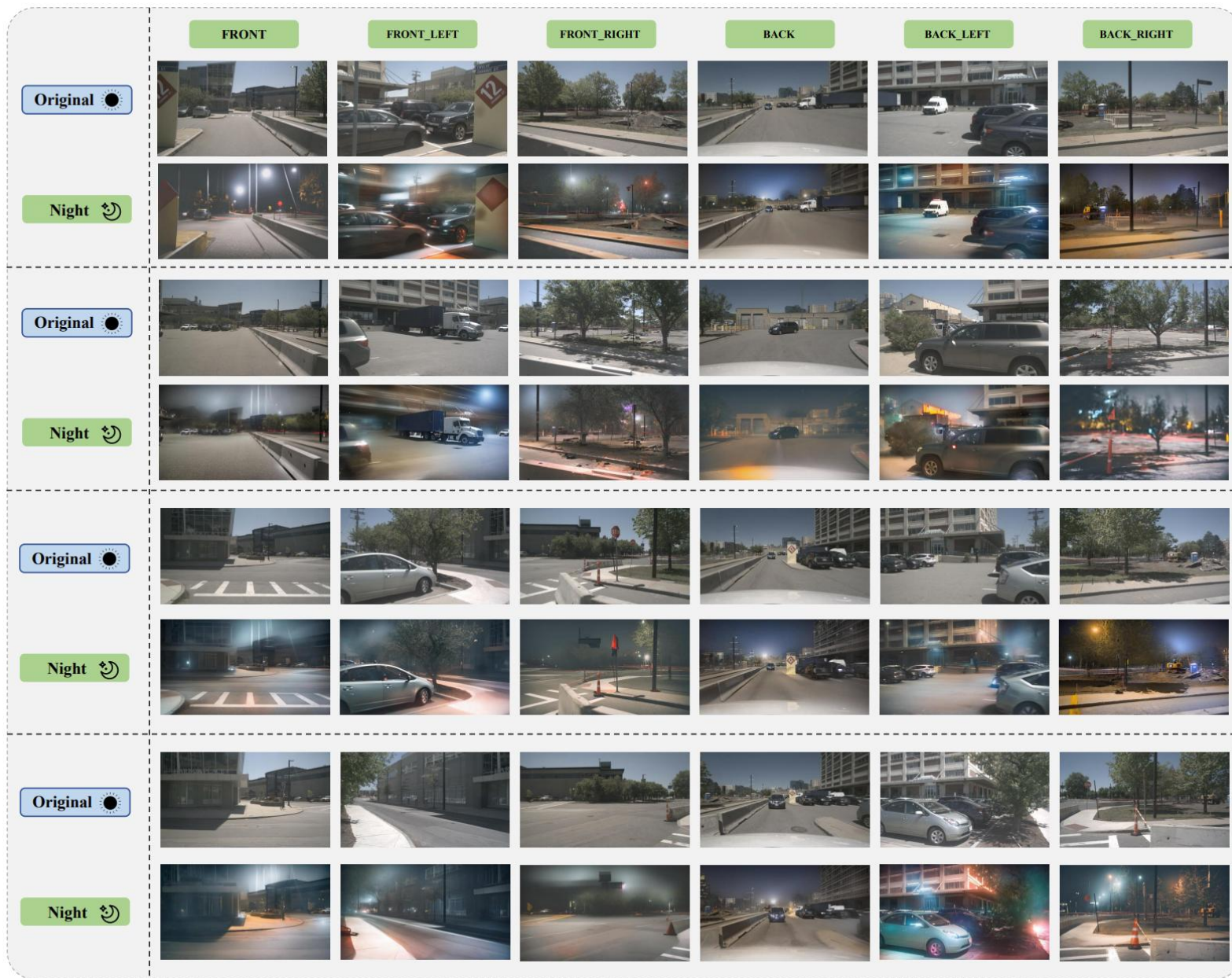
Monocular





Visualization

Multi-view



We also have the test-time solution!



Empowering Large Language Models with 3D Situation Awareness

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3D Scene Understanding Tasks:

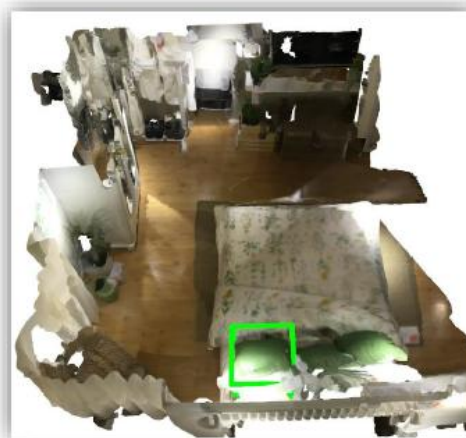
- Inputs: RGB-D scans and language instructions
- Outputs: 3D bounding boxes (3D visual grounding), captions, answers
- Application: Help the agent understand 3D world



- Driven by the great success of LLMs in image domain, their application in 3D scene understanding has emerged as a new trend.
- A key difference between 3D and 2D is that the situation of an egocentric observer in 3D scenes can change, resulting in different descriptions (e.g., “left” or “right”).



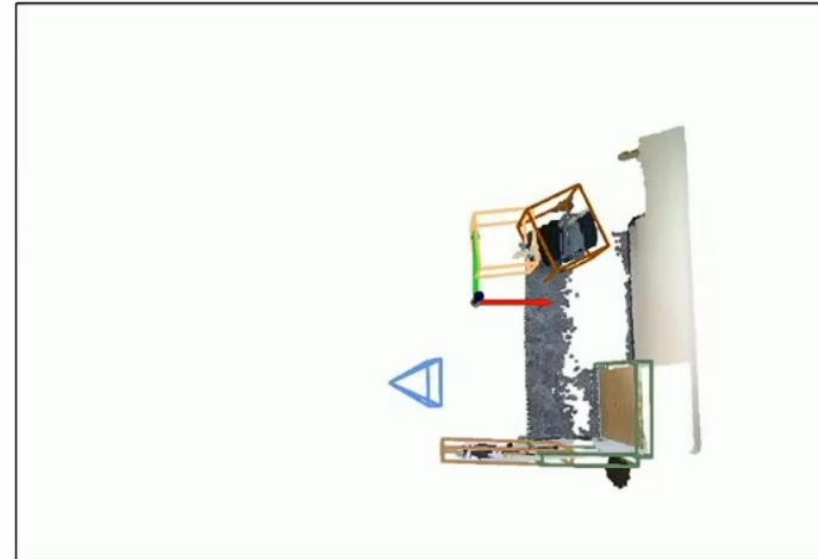
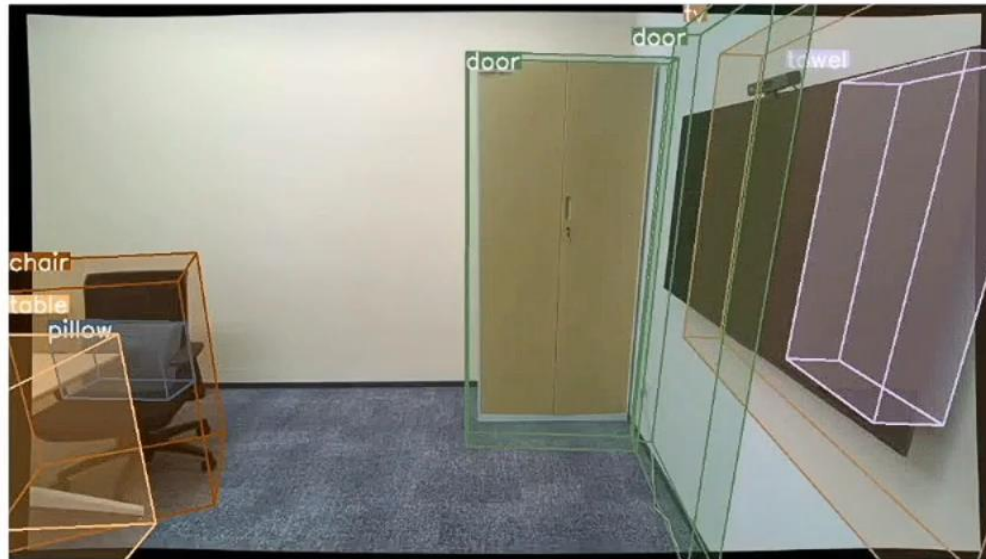
Front View



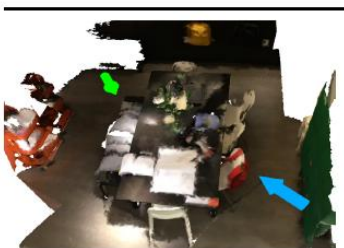
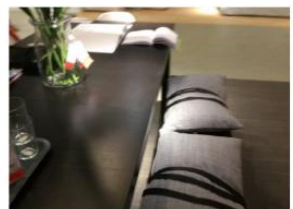
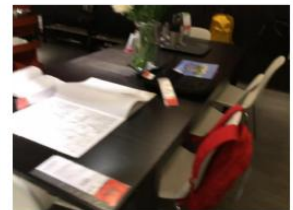
Back View

Query: “Facing the bed, the pillow on the far right.”

- However, current LLM-based methods overlook the egocentric perspective and use datasets from a global viewpoint.
- To address this issue, we propose a novel approach to automatically generate a situation-aware dataset by leveraging the scanning trajectory during data collection and utilizing Vision-Language Models (VLMs) to produce high-quality captions and question-answer pairs.



- **Automatic Data Generation.** Our data generation process leverages the natural exploratory behavior inherent in RGB-D videos, which serves as first-person navigations through 3D environments.
- **Dataset Verification and Refinement.** To ensure the quality and reliability of the generated dataset, we implement a verification and refinement process leveraging GPT-4.

	SceneVerse	
In this Dining Room scene, there are 5 chairs, 2 carts, 1 cabinet, and 3 backpacks. The cabinet is positioned to the right of the cart, while the cart is to the left of the cabinet. The chair is also to the left of the cabinet, and to the right of the cart. The objects in this room suggest a functional and organized space, suitable for dining or social gatherings. The chairs provide seating, the cabinet offers storage, and the carts can be used for serving or transportation.		
		Simple: I am in a modern living room. In front of me, there is a long black table with a clear vase of green plants and a stack of book on it. Detail: I am standing in a well-lit room with a modern interior. In front of me is a sleek, black rectangular table with a glass vase containing vibrant green plants on the left side. To the right of the table, there are two gray pillows with black stripes neatly placed against the wall. On the table, an open book lies flat, and in front of it, a clear glass with a green rim and a white tag stands next to another glass. To the left, there's a glass container with a green lid and a white price tag attached to it. The overall setting appears to be a modern living room or a showroom with a minimalist aesthetic.
		Simple: I am in a living room. There is a black table with papers and three chairs next it. Detail: I am in a room with a desk in the foreground. On the desk, there's a black surface with various items scattered on it, including papers, a red backpack hanging on a chair, and a plastic bag. To the left of the desk is a wooden shelf with some objects on it. In the background, there's a television set and a yellow bag on a stand. The room has a dark ambiance with some bright spots, possibly from a light source outside the frame.

EXAMPLES OF OUR VIEW2CAP SITUATION CAPTIONS AGAINST SCENEVERSE. WE MARK FACTS IN GREEN AND SPATIAL RELATIONS IN BLUE.

Method

- We introduce a situation grounding module to explicitly predict the position and orientation of the observer's viewpoint, thereby enabling LLMs to ground situation descriptions in 3D scenes.
- We treat each object as anchor we only need to predict the offset from the anchor position to the situation position and the angular difference between the anchor rotation and the situation

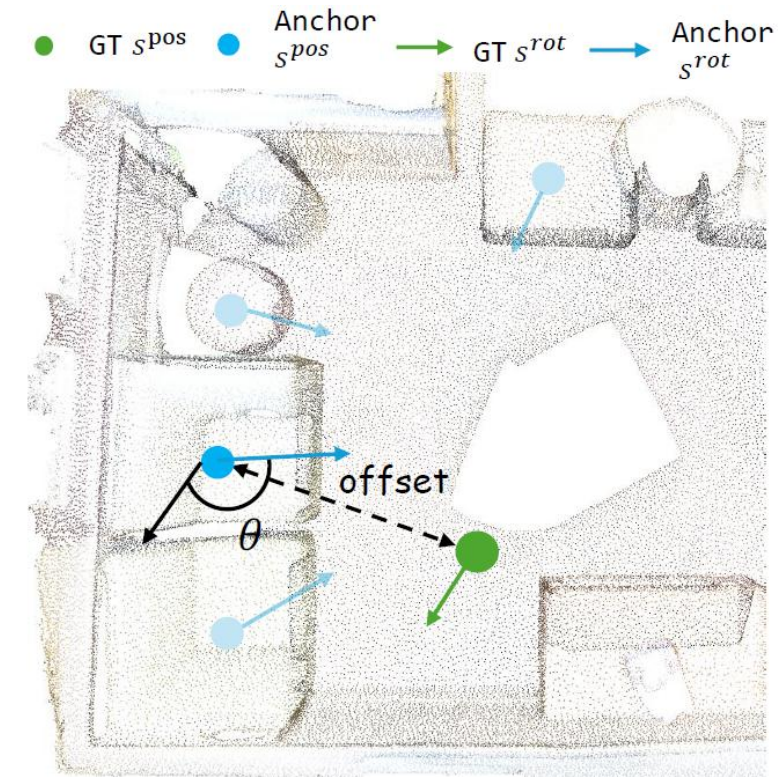
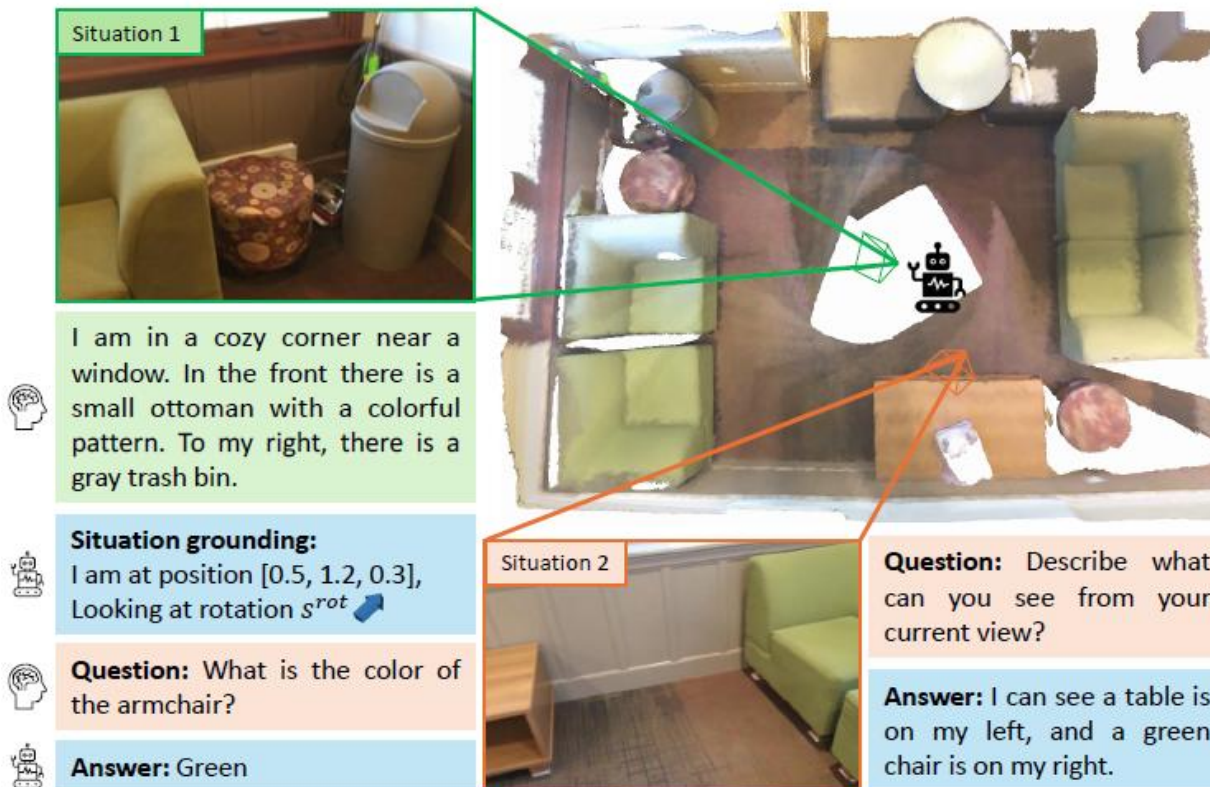
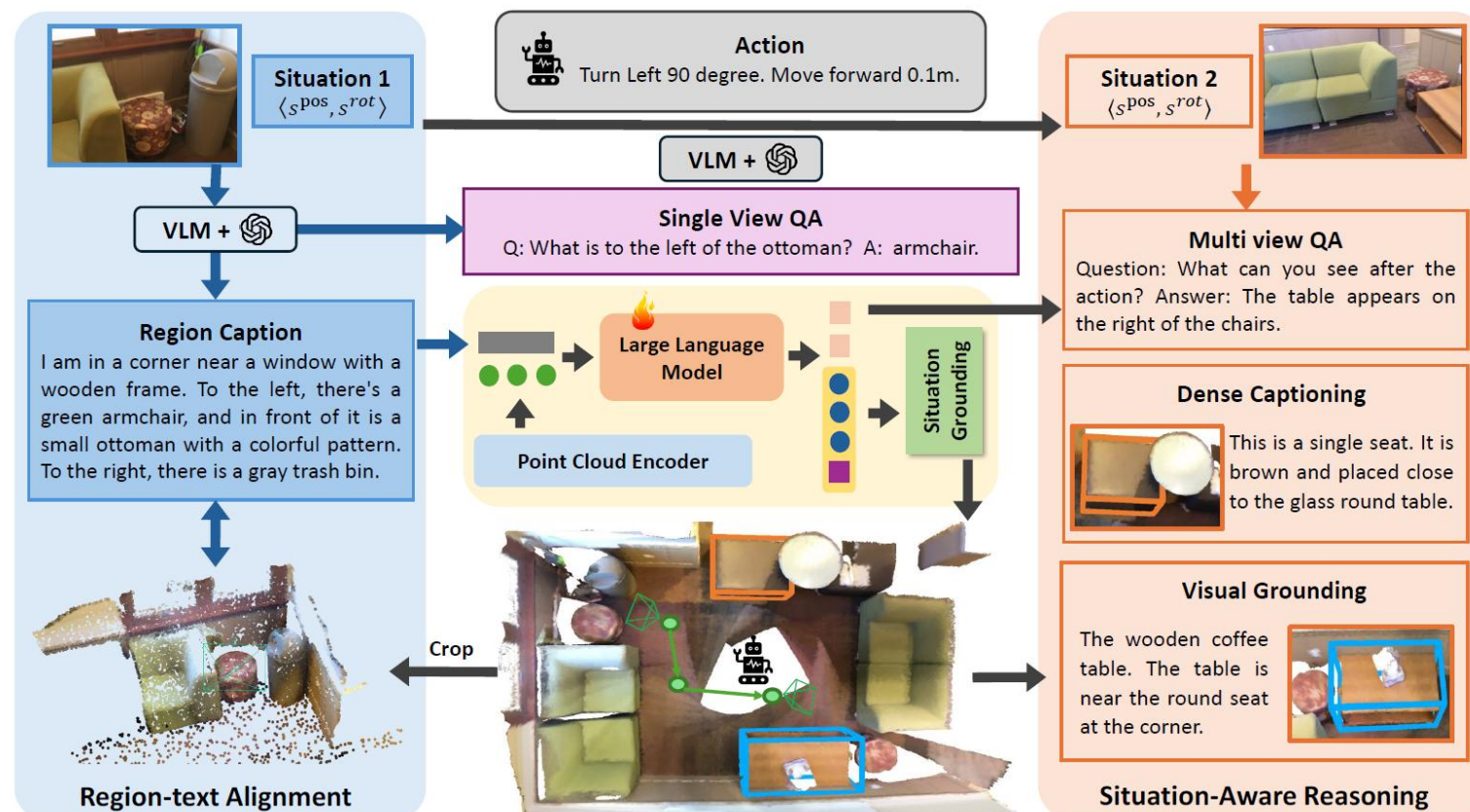


ILLUSTRATION OF 3D
SITUATION AWARENESS.

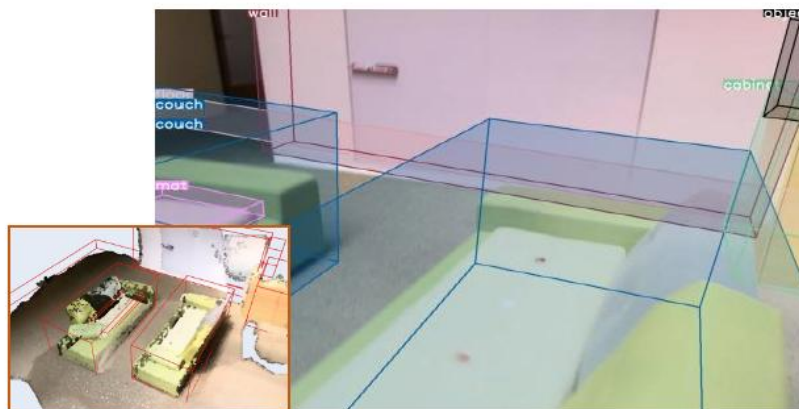
- The left part illustrates the process of region-text alignment. Paired point cloud and caption data are generated using VLM and RGB-D videos. The LLM is fine-tuned to align features from the point cloud encoder and generated region caption.
- The right part shows the situation-aware instruction tuning process, where QA data is generated using multi-view images and corresponding actions.



- We evaluate our method on three well-established 3D scene understanding tasks.
- Compared with LEO, our method surpasses 2.8 CIDEr scores on Scan2Cap and 4% on EM@1 on SQA3D.
- This improvement underscores the effectiveness of integrating 3D situational awareness into LLMs for enhancing 3Dscene understanding and reasoning capabilities.

Model	Scan2Cap (val)					ScanQA (val)					SQA3D (test)
	C	B-4	M	R	Sim	C	B-4	M	R	EM@1	EM@1
<i>Task-specific models</i>											
Scan2Cap [10]	35.2	22.4	21.4	43.5	-	-	-	-	-	-	41.0
3DJCG [3]	47.7	31.5	24.3	51.8	-	-	-	-	-	-	-
Vote2Cap-DETR [7]	61.8	34.5	26.2	54.4	-	-	-	-	-	-	-
ScanRefer+MCAN	-	-	-	-	-	55.4	7.9	11.5	30.0	18.6	-
ClipBERT	-	-	-	-	-	-	-	-	-	-	43.3
ScanQA [1]	-	-	-	-	-	64.9	10.1	13.1	33.3	21.1	47.2
SIG3D[27]	-	-	-	-	-	68.8	12.4	13.4	35.9	-	52.6
<i>Generalist models</i>											
3D-VisTA [46]	66.9	34.0	27.1	54.3	53.8	69.6	10.4	13.9	35.7	22.4	48.5
3D-LLM (FlanT5)	69.4	12.0	14.5	35.7	-	-	-	-	-	-	-
LL3DA [8]	65.2	36.8	26.0	55.1	-	76.8	13.5	15.9	37.3	-	-
LEO [18]	72.4	38.2	27.9	58.1	55.3	101.4	13.2	20.4	49.2	24.5 (47.6)	50.0 (52.4)
Ours	75.2	38.9	29.0	58.7	56.3	89.8	14.6	17.5	42.9	22.9 (40.2)	54.0 (56.0)

**COMPARISON WITH STATE OF THE ART MODELS ON
3D VL UNDERSTANDING TASKS.**



GT: *couch, mat, couch, cabinet*

Original: I am in a minimalist, modern room with a beige carpet. In the foreground, there's a low, padded **bench** in soft green and cream colors with small red dots. On the left, part of a similar **bench** is visible. In the background, a white wall has a small, rectangular **object** attached to it. **To the right, there's a wooden piece of furniture, possibly a cabinet or table.** The overall space is clean and uncluttered.

Refined: I am in a minimalist, modern room with a beige mat. In the foreground, there's a low, padded **couch** in soft green and cream colors with small red dots. On the left, part of a similar **couch** is visible. In the background, a white wall has a small, rectangular **cabinet** attached to it. The overall space is clean and uncluttered.



GT: *blackboard, picture, wall, computer, clock, monitor, chair, desk, monitor, shelf*

Original: I am in a room with a whiteboard on the left side and a wooden **door** on the right, with a clock hanging above the **door**. On the wall next to the clock is a poster of a blue character. In front of the door, there is a desk with a computer monitor, **speakers**, and various items on it.

Refined: I am in a room with a blackboard on the left side and a **wall** on the right, with a clock hanging above the **wall**. On the wall next to the clock is a picture. In front of the wall, there is a desk with a computer monitor.

Empowering Large Language Models with 3D Situation Awareness:

- We introduce View2Cap, a scalable 3D dataset that provides paired data of situational positions, rotations, region point clouds, textual descriptions, and QA pairs. This dataset is generated automatically without the need for 3D labels or extensive manual annotations.
- We propose the Situation Grounding SG module, that can be integrated into existing LLMs, allowing for explicit grounding of situational descriptions in 3D scenes. This module transforms pose estimation into a classification problem, facilitating easier training and improved spatial reasoning.
- Through extensive experiments, we demonstrate that combining our situational dataset and grounding module significantly enhances the situational awareness and performance of existing 3D LLMs in various tasks.



CLEA: Closed-Loop Embodied Agent for Enhancing Task Execution in Dynamic Environments

Mingcong Lei*, Ge Wang*, Yiming Zhao, Zhixin Mai, Qing Zhao, Yao Guo, Zhen Li, Shuguang Cui, Yatong Hant, Jinke Rent

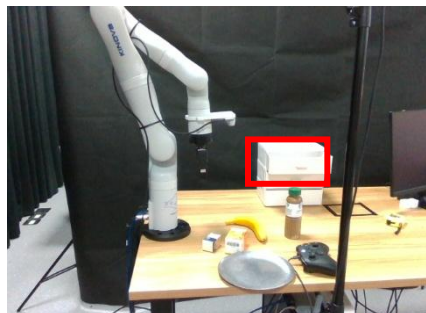


Motivations

The difficulties of deploying Embodied AI to the Real World: The environment is

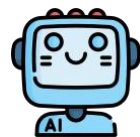
- partially observed
- dynamic

Thus, it is not a simple “predict the next token” task for Multi-model Large Language Model (MLLM).

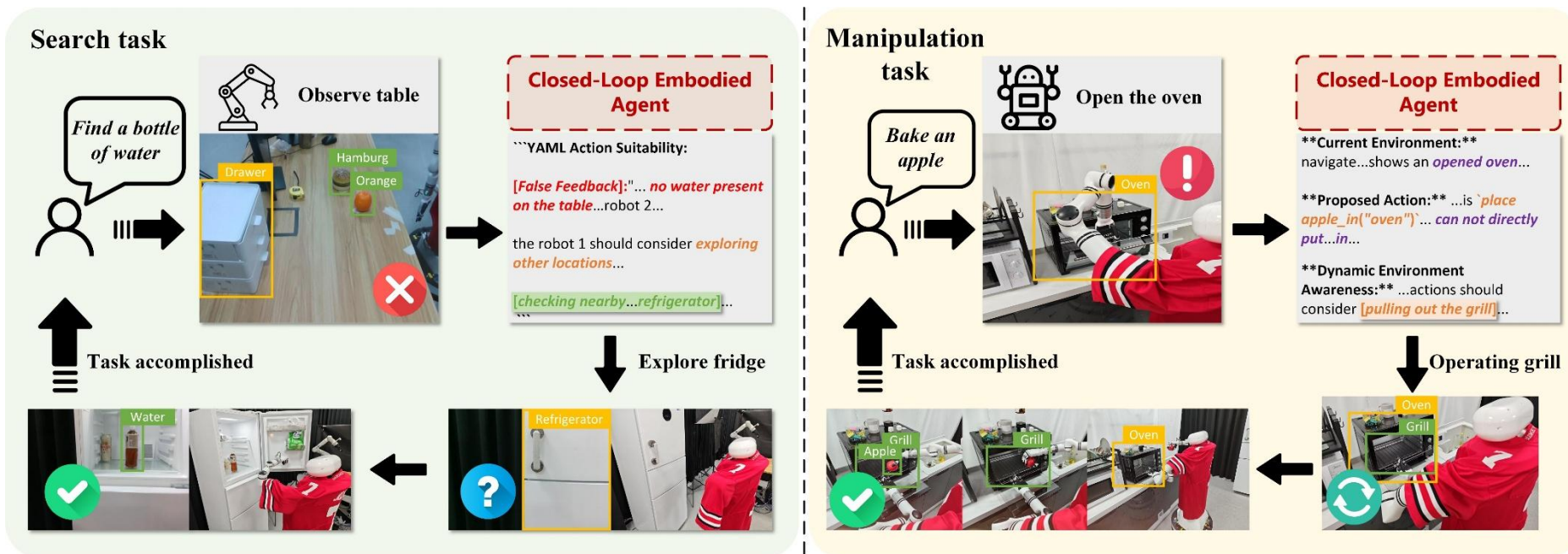


Task: Pick up pills

Action: Open the upper drawer

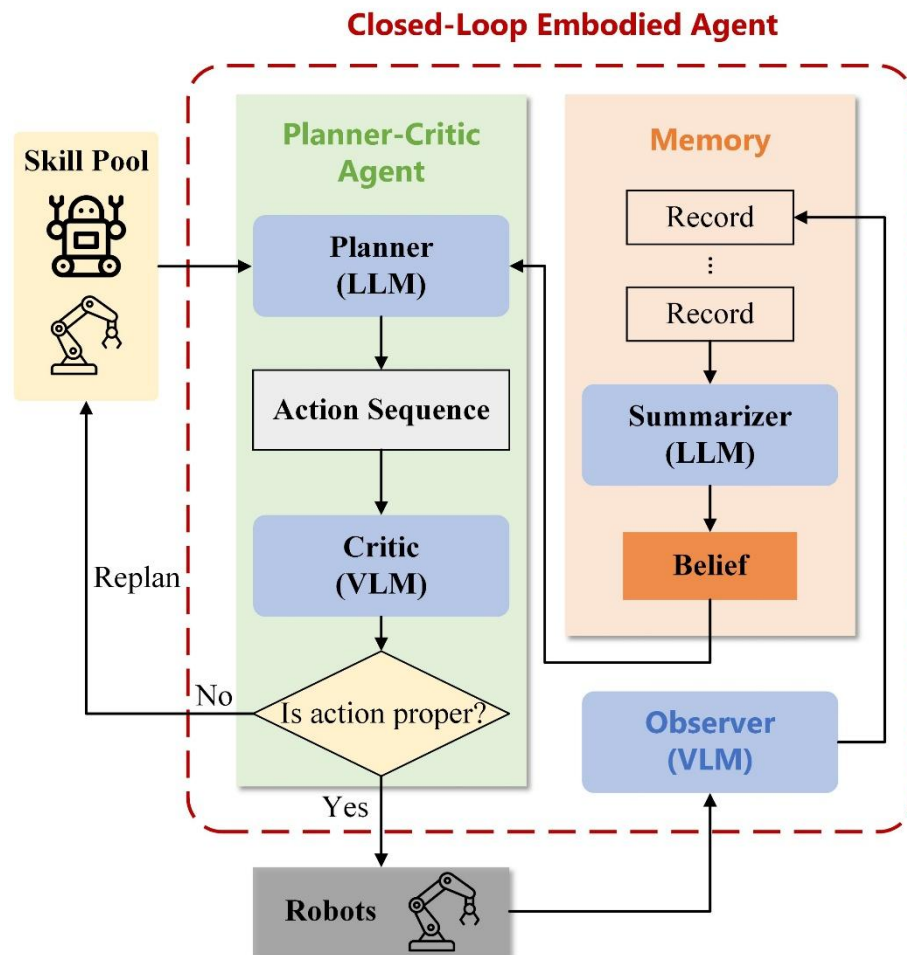


Contributions



- We proposed a **closed-loop planning** algorithm that is suitable for real-world planning.
- We deploy open-source MLLM-Agent in a **real environment** for dynamic tasks (needs to 'find' objects).

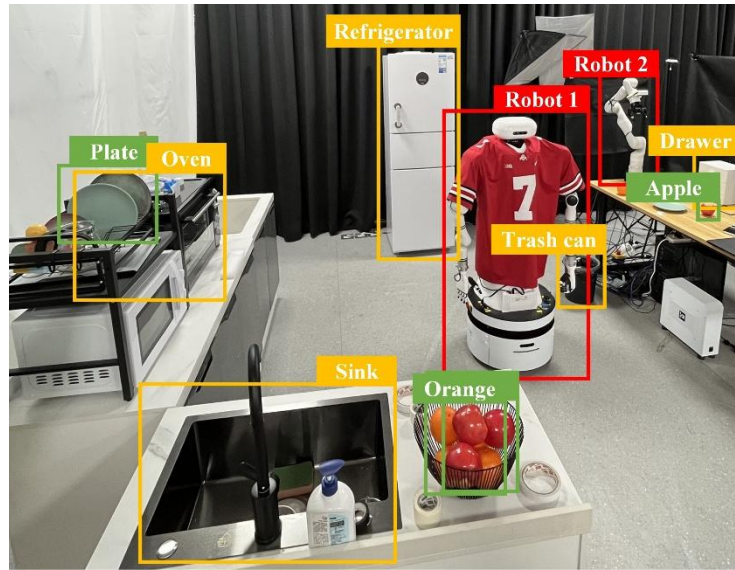
Methodology



Key components:

- Closed Loop Planner
- Simple Sequential Memory

Experiments

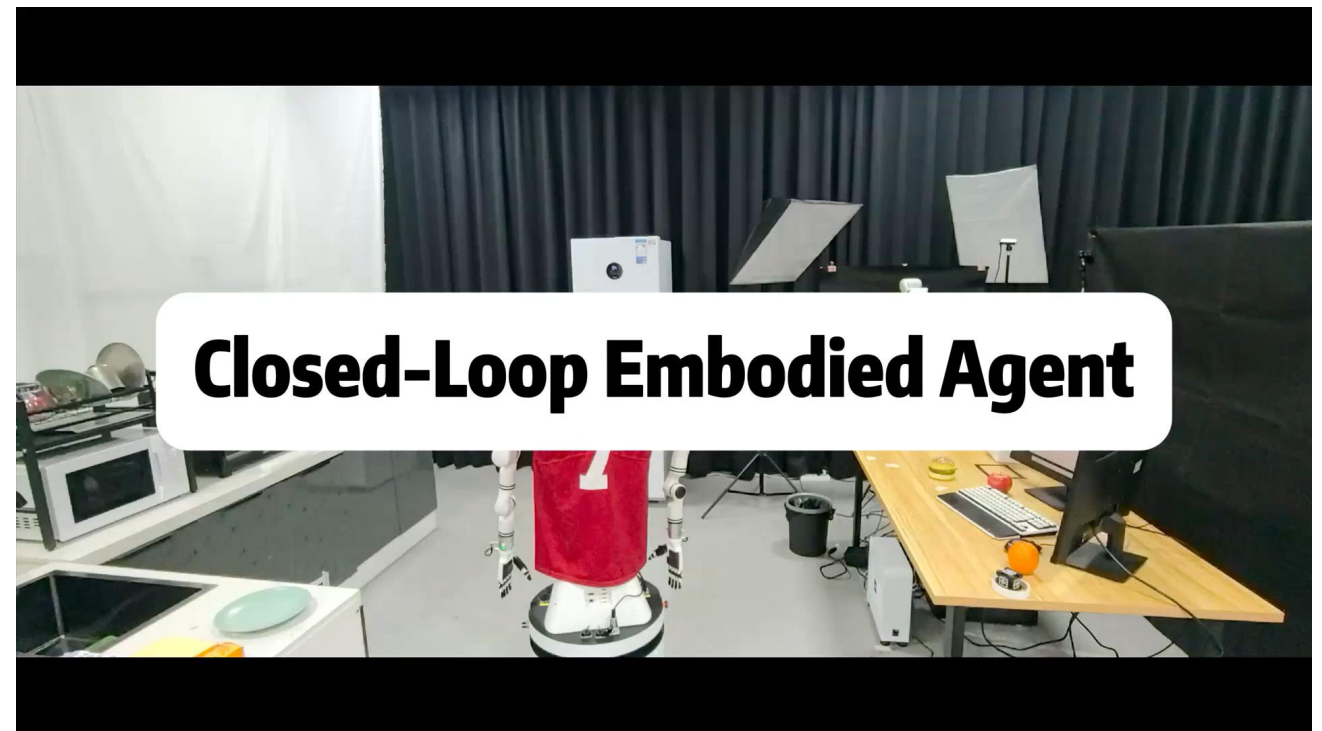
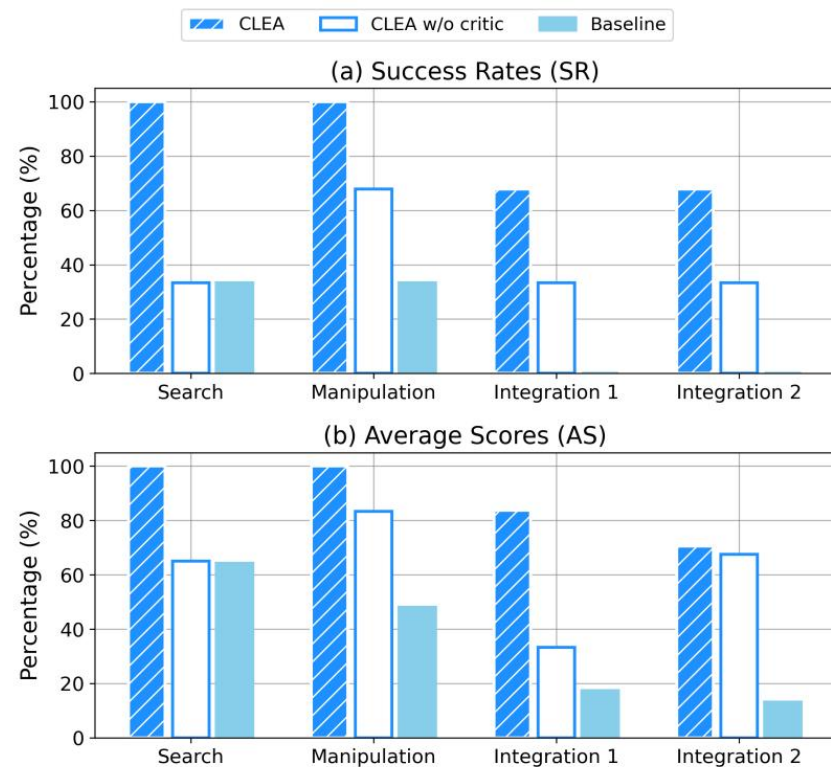


PREDEFINED SKILL POOL IN THE ENVIRONMENT

Skill pool	Description
open(robot, openable_object)	robot open object
close(robot, openable_object)	robot close object
pick_from(robot, object, space)	robot pick object from space
release_to(robot, space)	robot release the object on its hand to space
go_to(robot, navi_point)	robot navigate to navigation point

- We deploy our agent in the real world in a typical kitchen environment with two robots.
- The agent should utilize two robots to complete long-term tasks.
- We design a skill pool that is suitable for most tasks in a kitchen environment.
- We use **SLAM** for the navigation skill, and **Inverse kinematics** to implement operational skills.

Experiments



contact



Take aways

- Pre-training is still useful for feature representation. However, which type of feature representation should we design? Still Open Issue, 3D/4D?
- Perception and reasoning for planning should have a trade-off, especially for open world driving.
- Generalization and robustness are not only for OOD, but also for unexpected corner cases.
- World model is not only for simulation, but also with world priors such as environment geometry, traffic priors and physical interactions.



Acknowledgement



Xiaoyu Ye



Hongbin Lin



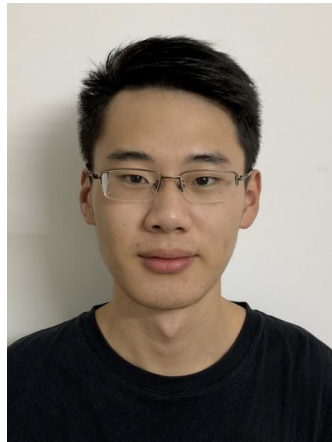
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Deep Bit Lab

Thanks for Listening !

Q&A

Zhen Li